A Reinforcement Learning-based Adaptive EWMA Control Chart: Simulation and Bus Ridership Validation

Usman Ibrahim[a,b,*], Mohammad AlDurgam[a,c], Nasir Abbas[d], Alaa Khamis[a,b]

[a] *Industrial and Systems Engineering Department, King Fahd University of Petroleum & Minerals, Dhahran, 31261, Saudi Arabia*
[b] *AI for Smart Mobility Lab, IRC for Smart Mobility and Logistics, King Fahd University of Petroleum & Minerals, Dhahran, 31261, Saudi Arabia*
[c] *IRC for Smart Mobility and Logistics, King Fahd University of Petroleum & Minerals, Dhahran, 31261, Saudi Arabia*
[d] *Department of Mathematics, King Fahd University of Petroleum & Minerals, Dhahran, 31261, Saudi Arabia*

Abstract

Several efforts have been made to improve the performance of control charts to enhance their detection capabilities for better process management. In this study, we propose an EWMA control chart based on reinforcement learning for process monitoring. We formulate the monitoring task as a Markov Decision Process and train an agent using Proximal Policy Optimization to learn an optimal signaling rule. The performance of the chart is evaluated using the average run length and expected quadratic loss over a range of shift magnitudes. The numerical results show that the proposed chart improves the detection of small and moderate shifts compared to the classical methods while maintaining comparable performance for large shifts. We further illustrate the real applicability of the approach through a transportation case study using data from the Chicago Transit Authority. The proposed chart successfully identifies change points around major disruptions such as the 2007–2008 economic crises and the 2020 COVID-19 pandemic.

Keywords:
Reinforcement Learning; Control Chart; EWMA; Transport Ridership; Proximal Policy Optimization; MDP

1. Introduction

Maintaining a high quality product and process in modern manufacturing and service industries is very important in order to meet customer expectations and minimize production costs (Montgomery, 2020). As systems become automated, systematic tools for monitoring variations are essential to achieve sustainable competitiveness. In transportation systems, demand data often have seasonal variation and long term trends driven by factors such as weather, service changes, and socioeconomic cycles (Hightower et al., 2024). These characteristics can present challenges for monitoring approaches that assume a stable distribution or fixed control limits over time. As a result, adaptive and data-driven monitoring schemes that accommodate evolving behavior and contextual variability can provide more interpretable and relevant alerts in practice (Zaman et al., 2025).

Process variation can be broadly classified into two types based on their causes. These are natural causes and assignable causes (Colosimo et al., 2024). Natural variation arises from inherent randomness in the process, such as environmental fluctuations or measurement noise, and usually does not significantly affect the quality of the process. On the other hand, assignable causes arise from specific identifiable factors, such as equipment failures, raw material defects, or human errors. These shifts in process behavior can lead to degradation in quality of products and services if not detected and corrected promptly. A fundamental objective of quality engineering, therefore, is to distinguish between these two types of variation and respond appropriately to prevent the propagation of faults or frequent unnecessary stoppage of the process.

Statistical Process Control (SPC) is a set of statistical tools that can be used to detect assignable causes by using statistical schemes, such as control charts. Classical control charts include Shewhart (Shewhart, 1930), cumulative sum (CUSUM) (Page, 1954), and exponentially weighted moving average (EWMA) charts

*Corresponding author
Email address: g202210480@kfupm.edu.sa (Usman Ibrahim)

(Roberts, 1959). The charts are used to monitor process statistics over time and raise alarms when significant deviations from the nominal values occur (Aykroyd et al., 2019). These charts are designed to balance the trade-off between false alarms, i.e, signaling during natural variation, and detection delay when actual process shift occurs. To ground the discussion, Figure 1 illustrates a classical Shewhart control chart under three regimes. In the in-control (IC) state, the process fluctuates around the target value and remains within the predefined upper and lower control limits (UCL and LCL). In contrast, for the second and third regimes, the presence of assignable cause has induced an out-of-control (OOC) behavior, which is detected when the plotted statistic exceeds the control limits. Although effective under normal conditions, such classical approaches may not fully exploit statistical trends in the data or adapt to evolving dynamics of the monitored process. In this study, we introduce a reinforcement learning approach that directly learns from sample observations and EWMA statistics to detect OOC conditions. This method offers a more flexible and potentially more responsive alternative to traditional control charts.

Studies (Kazmi and Noor-ul Amin, 2025; Abbas et al., 2025; Chang and Park, 2025; Ajab et al., 2025; Li et al., 2024; Yeganeh et al., 2024; Acosta and Oliveira Sant'Anna, 2023) have explored the integration of machine learning into SPC, which enables adaptive decision-making through interaction with the process environment. RL is a good approach to learn optimal monitoring decision in settings where the system dynamics are complex or partially unknown. By formulating SPC as a sequential decision-making problem, RL-based controllers can dynamically adjust thresholds, sampling policies, or chart parameters to improve monitoring performance (Viharos and Jakab, 2021).

Our Contribution

We developed an RL-based EWMA control chart to improve univariate process monitoring. We train a signaling agent using standard Proximal Policy Optimization (PPO) (Schulman et al., 2017) to learn a monitoring policy that balances detection speed with false alarms. The key idea is to allow the agent to learn a nonlinear mapping from sample and EWMA statistics to signal decisions without the use of the predefined control limits as in classical charts. Our main contributions are summarized as follows:

- We formulate the process monitoring problem as a sequential stopping decision and train a PPO policy that maps the EWMA state to a signal/continue action under a reward structure that trades off detection delay and false alarms.
- We propose a post-training threshold calibration step that enforces a target performance without constraining policy learning. This aligns the final operating point with the classical control limit tuning.
- The resulting RL-based EWMA chart achieves better OOC detection performance than the existing classical charts.

This approach combines the adaptability and modeling flexibility of RL with the statistical rigor of control chart methods. The proposed chart offers a hybrid solution that achieves better overall performance in process monitoring.

The remainder of the article is organized as follows: Section 2 highlights some of the previous literature on control charts, RL, and recent integrations of the two concepts. In Section 3, we present a detailed description of the proposed chart, including the formulation of the monitoring problem as an MDP and the training setup. The simulation results and the comparative analysis with existing charts are presented in Section 4. We discuss additional considerations, such as the effect of parameter estimation in Section 5. Section 6 presents the case study using bus ridership, while Section 7 concludes the study and discusses future research directions.

2. Literature Review

2.1. Control Charts

SPC methods largely rely on control charts to monitor process behavior and detect abnormal deviations. A basic control chart is the Shewhart chart, which uses statistical limits to signal when a process metric (e.g. the sample mean or range) crosses a fixed threshold. Shewhart charts are simple to construct and detect large process shifts quickly, but they are less sensitive to smaller or gradual shifts (Montgomery, 2020). To improve sensitivity for detecting small sustained shifts, CUSUM EWMA charts are widely used.

The CUSUM chart accumulates deviations of the process statistic from its target over time and signals an OOC condition when the cumulative sum exceeds a decision threshold. By compounding deviations, the CUSUM chart can detect small changes in the process much faster than a Shewhart chart (Abbas,

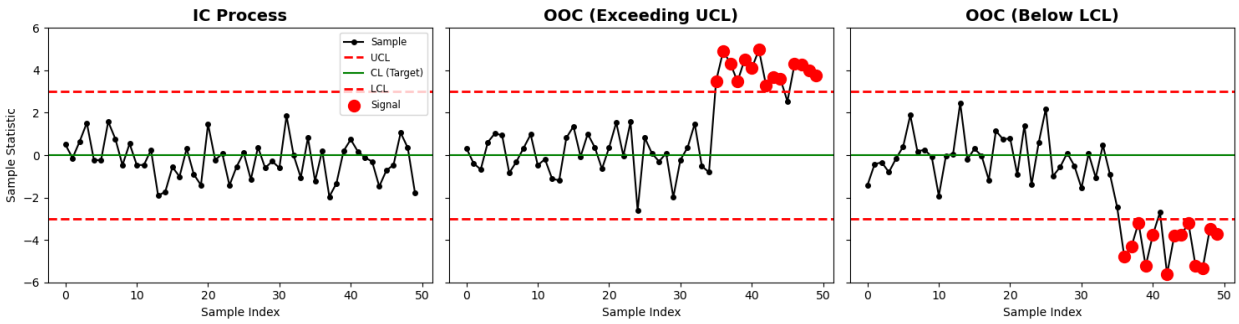


Figure 1: Example control chart behavior under three conditions: an IC process with observations within the control limits, an OOC upward shift producing signals above the UCL, and an OOC downward shift producing signals below the LCL (red markers).

2023). CUSUM methods are closely related to sequential likelihood ratio tests for change detection, and studies have shown that CUSUM can be derived as optimal or near optimal procedures under certain assumptions (Tsui et al., 2012). The EWMA reacts to small changes by smoothing out noise and giving more influence to the most recent data, which also yields faster detection of small changes compared to Shewhart charts (Ghasemian and Noorossana, 2024). Both CUSUM and EWMA charts require design parameters, for example, the reference value or weighting factor, that determine their sensitivity. These are typically set based on statistical considerations and the magnitude of shifts of interest. A comparison of classical control charts including their plotting statistics, control limits, and design parameters is summarized in Table 1.

There are also control charts for multivariate data, these include the Hotelling's T^2, multivariate CUSUM, and multivariate EWMA and their variants (Bersimis et al., 2007; Ali et al., 2024; Riaz et al., 2025; Nishad and Varadharajan, 2025). Together with their univariate counterparts, these charts have been used to monitor and control process stability in manufacturing, healthcare, and other domains. However, classical control charts are generally designed based on static parameters, such as fixed control limits and smoothing factors. In modern, rapidly evolving applications, such approaches lose their efficiency as operating conditions change. This limitation has motivated research into more adaptive (see (Psarakis, 2015)), data-driven monitoring approaches, including the use of RL to improve conventional control chart methodologies.

2.2. *Reinforcement Learning*

Reinforcement learning (RL) is an area of machine learning that aims to train autonomous agents to make sequential decisions through trial and error interactions with an environment (Sutton et al., 2018). In the standard RL framework, an agent observes the current state of the environment, takes an action, and receives a reward signal that reflects immediate feedback for its action. The environment then transitions to a new state, according to the underlying dynamics. This process is repeated until the agent learns a policy, defined as a mapping from states to actions, that maximizes long-term cumulative reward (Bai, 2025).

Sequential decision-making is modeled as a Markov Decision Process (MDP), characterized by the state space, action space, reward function, and state transition probabilities. RL algorithms can be broadly categorized into value-based methods, policy-based methods, and actor–critic methods (Sutton et al., 2018). Value-based methods, such as Q-learning, focus on learning a value function that estimates the expected cumulative reward for each state-action pair. Decisions are made by choosing actions with maximum value estimates. Policy-based methods directly update the policy on expected reward, and actor–critic approaches combine both techniques by having an explicit policy (called actor) informed by a value function (called critic). Modern advances in RL often involve deep RL, where deep neural networks represent the value function or policy (Li et al., 2018). This has enabled RL agents to handle high dimensional state and action spaces and to achieve human level or better performance in domains such as games and robotics (Mnih et al., 2015).

A key feature of RL is the tradeoff between exploration and exploitation. The agent must balance exploiting current knowledge to gain rewards with exploring new actions that might yield higher rewards in the future. Through repeated interaction with the environment, the agent iteratively improves its policy and learns from experience. For a detailed discussion of the RL algorithms, the reader is referred to Shakya et al. (2023).

Table 1: Basic control charts and their design parameters.

Chart	Control Statistic	Control Limits	Design Parameters
Shewhart(Shewhart, 1930)	$X_t/\bar{X}$	$UCL = \mu_0 + L\sigma,\ LCL = \mu_0 - L\sigma$	L set to achieve ARL_0 (e.g., 370)
CUSUM(Page, 1954)	$C_t^+ = \max(0, C_{t-1}^+ + X_t - k)$ $C_t^- = \max(0, C_{t-1}^- - X_t + k)$	$C_t > H$	k, H tuned to ARL_0
EWMA(Roberts, 1959)	$Z_t = \lambda X_t + (1-\lambda)Z_{t-1}$	$UCL = \mu_0 + L\sigma\sqrt{\frac{\lambda}{2-\lambda}}$ $LCL = \mu_0 - L\sigma\sqrt{\frac{\lambda}{2-\lambda}}$	$\lambda \in (0,1]$, L set for ARL_0
AEWMA (Capizzi and Masarotto, 2003)	$x_t = x_{t-1} + \varphi(e_t),\ e_t = y_t - x_{t-1},\ x_0 = \mu_0$	$\lvert x_t - \mu_0\rvert > h$	$\varphi(\cdot)$ and its parameters, h

In the context of process control and monitoring, an RL agent can be seen as continuously learning the best control or monitoring actions to take in response to process observations. Unlike supervised learning, RL does not require labeled examples of faults or normal operations. Instead, it learns an optimal policy based on reward feedback, for example, rewards that encourage quick and accurate detection of anomalies. This ability to learn adaptively and optimize decisions makes RL a promising paradigm for enhancing the abilities of control charts in process monitoring.

2.3. RL–based Control Charts

Integrating RL into SPC has opened up new avenues for more adaptive and intelligent process monitoring methods. Li et al. (2024) formulated the monitoring task as a sequential decision-making problem that an RL agent can learn to solve using Q-learning with a nonparametric CUSUM chart. Their method involves selecting the most informative sensor to sample at each time epoch and uses the CUSUM statistic as part of the system state. It was shown that the RL-based scheme achieved faster detection of mean shifts in high dimensional streaming data compared to classical fixed sampling techniques.

Another study by Chang and Park (2025) proposed an RL-based real-time contrasts (RTC) control chart. In the study, an RL agent is trained to decide the size of the moving window (the amount of recent data used to compute the contrast statistic). They used the current process data stream as the state of the environment, and the action is to increase or decrease the moving window length for the RTC chart before the next sample. This RL-based control chart was shown to detect process shifts more quickly than a conventional RTC chart with a fixed window (Chang and Park, 2025). The main advantage in such an approach is that the control chart no longer has static settings, rather it becomes a self-learning system that can adapt its sensitivity based on process history. Over time, the RL agent will learn the optimal chart parameters or sampling decisions for different situations, something that would be difficult to achieve with manual tuning or predefined rules.

These few examples available in the literature highlight how we can exploit the power of RL techniques to improve the performance of control charts. This paradigm shift allows control charts to become proactive by dynamically adjusting their parameters as the process evolves. It should also be noted that this is a relatively new paradigm in control charts designs, and so far there is a dearth of studies in the area.

2.4. RL for Anomaly Detection

Beyond direct augmentation of control charts, RL has also been applied to anomaly detection tasks. In anomaly detection, the objective of the agent is to identify unusual patterns or faults in a system as early and accurately as possible. An RL agent can be trained to make these decisions by rewarding correct and timely detections and penalizing missed detections and false alarms.

One of the approaches for RL-based anomaly detection treats the problem as a form of partially observable control. For example, Zhang and Hoi (2019) model sequential change detection in a multi-sensor system as a combinatorial multi-armed bandit problem. The agent selects the sensors to query at each time, in order to detect a change while limiting the observations. Also, deep RL methods have been explored for a variety of anomaly detection applications ranging from network intrusion detection to fault diagnosis in engineering systems (Yang et al., 2024; Zha et al., 2020; Elaziz et al., 2023). A systematic review by Arshad et al. (2022) surveyed several deep RL models for anomaly detection (2017–2022) and found that in many cases RL-based approaches have better performance compared to other methods.

Although this intersection of RL and SPC is still emerging, the early results are promising. They indi-

cate that by harnessing the power of RL, we can have more agile and context-aware methods that that improve detection rates and reduce downtime in industrial operations (Arshad et al., 2022). Based on previous discussions, it is clear that the convergence of RL with statistical process monitoring represents a novel direction for quality control research. The RL provides a framework for continual learning and decision optimization that can address some limitations of classical SPC, such as the need for manual tuning and the difficulty of handling evolving process behaviors.

3. Methodology

3.1. The PPO Algorithm

PPO is an on-policy policy improvement method that aims to update a stochastic policy while keeping successive policies close, so that learning is stable and expected return improves reliably (Schulman et al., 2017). A common starting point is a lower bound on the performance change between a current policy π_k and a candidate policy π, written as a surrogate improvement term minus a penalty proportional to the expected total variation (TV) distance between the two policies (Achiam et al., 2017; Murphy, 2024). Let $p^{\gamma}_{\pi_k}(s)$ denote the normalized discounted state-visitation distribution under π_k, and let $A_{\pi_k}(s,a) = Q_{\pi_k}(s,a) - V_{\pi_k}(s)$ be the advantage. With $\mathrm{TV}(p,q) \triangleq \frac{1}{2}\|p-q\|_1$, one form of the bound is

$$J(\pi) - J(\pi_k) \;\geq\; \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim p^{\gamma}_{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\frac{\pi(a|s)}{\pi_k(a|s)}A_{\pi_k}(s,a)\right] - \kappa\,\mathbb{E}_{s\sim p^{\gamma}_{\pi_k}}\left[\mathrm{TV}(\pi(\cdot|s),\pi_k(\cdot|s))\right]. \tag{1}$$

for an appropriate constant κ. This motivates trust region updates (as in TRPO Schulman et al. (2015a)) that constrain the average divergence between policies (Murphy, 2024).

PPO replaces the explicit trust region constraint with a clipped surrogate objective (Schulman et al., 2017). Using the likelihood ratio

$$\rho_k(s,a) = \frac{\pi(a|s)}{\pi_k(a|s)}, \qquad \tilde{\rho}_k(s,a) = \mathrm{clip}(\rho_k(s,a), 1-\epsilon, 1+\epsilon),$$

(where $\mathrm{clip}(x,l,u) = \min(\max(x,l),u)$), PPO updates the policy by maximizing

$$\mathbb{E}_{(s,a)\sim p^{\gamma}_{\pi_k}}\Big[\,\min\big(\rho_k(s,a)A_{\pi_k}(s,a),\ \tilde{\rho}_k(s,a)A_{\pi_k}(s,a)\big)\Big]. \tag{2}$$

The clipping prevents overly large changes in action probabilities from dominating the update. An equivalent case form makes this explicit as

$$\min(\rho A, \mathrm{clip}(\rho)A) = \min(\rho A,\ g(\epsilon, A)),$$
$$g(\epsilon, A) = \begin{cases}(1+\epsilon)A, & A \geq 0\\ (1-\epsilon)A, & A < 0\end{cases}$$

so, positive advantage actions cannot be increased by more than a factor $(1+\epsilon)$ and negative advantage actions cannot be decreased by more than a factor $(1-\epsilon)$ in the surrogate objective (Murphy, 2024).

In practice, PPO is implemented with an actor–critic and advantages computed using Generalized Advantage Estimation (GAE), see Algorithm 1. The training loss combines (i) a value-regression term, (ii) the clipped policy term in (2), and (iii) an entropy bonus to discourage premature collapse to near deterministic policies (Schulman et al., 2015b, 2017; Murphy, 2024)

Algorithm 1: PPO with GAE

parameters θ, ϕ and environment state s
initialization
for iteration $k = 1, 2, \ldots$ **do**
  $\theta_{\text{old}} \leftarrow \theta$
  $(s_1, a_1, r_1, \ldots, s_T) \leftarrow \textsc{rollout}(s, \pi_{\theta_{\text{old}}})$
  $v_t \leftarrow V_\phi(s_t)$ **for** $t = 1{:}T$
  $(A_{1:T}, y_{1:T}) \leftarrow \mathrm{GAE}(r_{1:T}, v_{1:T}; \gamma, \lambda)$
  for $m = 1{:}M$ **do**
    $\rho_t \leftarrow \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$ **for** $t = 1{:}T$
    $\tilde{\rho}_t \leftarrow \textsc{clip}(\rho_t)$ **for** $t = 1{:}T$
    $L(\theta,\phi) \leftarrow \frac{1}{T}\sum_{t=1}^{T}\Big[\lambda_{\text{TD}}\,\textsc{clip}((V_\phi(s_t) - y_t)^2) - \lambda_{\text{PG}}\min(\rho_t A_t, \tilde{\rho}_t A_t) - \lambda_{\text{ent}}\,H(\pi_\theta(\cdot \mid s_t))\Big]$
    $\theta \leftarrow \theta - \eta\nabla_\theta L(\theta,\phi)$
  end for
end for

3.2. The Proposed RL-based EWMA Control Chart

The RL-based design of the EWMA control chart is modeled as an MDP. The underlying process produces observations X_t and is monitored using an EWMA statistic. The agent observes the sample value and EWMA statistic and decides whether to continue sampling or to signal. Since the true IC or OOC status is not directly observed, the problem can be viewed as partially observable MDP (POMDP), with the binary regime indicator acting as a hidden state. However,

following Li et al. (2024) we do not maintain an explicit belief state as in POMDP methods. Instead, we model the decision process on the observable monitoring statistics and learn a stopping rule on this information. The key notations used throughout the modeling and evaluation of the RL-based control chart are summarized in Table 2.

3.3. State, Actions, and Rewards

The observable state at time t is the raw sample observation X_t and the EWMA statistic. It should be noted that alternative formulation with only the EWMA statistic results in similar performance.

$$Z_t = \lambda X_t + (1-\lambda)Z_{t-1}, \qquad Z_0 = 0, \quad 0 < \lambda < 1, \tag{3}$$

where $X_t \sim \mathcal{N}(\mu_t, \sigma^2)$ is the process sample.

$$S_t = \{X_t,\ Z_t\} \in \mathbb{R}^2. \tag{4}$$

The environment maintains an unobserved binary indicator

$$H_t \in \{0, 1\}, \tag{5}$$

where $H_t = 0$ denotes the IC regime with mean $\mu_t = \mu_0$ and $H_t = 1$ denotes the OOC regime with mean $\mu_t = \mu_0 + \delta\sigma$. Once a shift occurs, the process remains in the OOC (absorbing state), until the agent signals and the episode ends. At each time t, the agent chooses an action

$$A_t \in \{0, 1\}, \tag{6}$$

where $A_t = 0$ (wait): continue sampling, and $A_t = 1$ (signal): declare an OOC condition and stop.

Shift occurrences are modeled using a geometric mechanism. While the process is in control ($H_t = 0$), a shift arrives with probability

$$\mathbb{P}(H_{t+1} = 1 \mid H_t = 0) = p, \tag{7}$$

The value of p should be selected in such a way that the agent sees enough IC as well as OOC episodes to learn a meaningful policy. The EWMA and sample values then induce the transition probabilities for the observable state (X_t, Z_t).

The one-step reward $R(S_t, A_t, H_t)$ is as follows:

$$R_t = \begin{cases} -c_s, & A_t = 0,\ H_t = 0, \\ -c_s - c_d, & A_t = 0,\ H_t = 1, \\ R_{\text{sig}}, & A_t = 1,\ H_t = 1, \\ -P_{\text{FA}}, & A_t = 1,\ H_t = 0, \end{cases} \tag{8}$$

where c_s is the sampling cost, c_d is the cost of operating in a defective state, R_{sig} is the reward for a correct signal, and P_{FA} is the penalty for a false alarm.

Figure 2 shows the RL control loop used in this study. At each time t, the PPO agent observes S_t, selects an action A_t, and the control chart environment returns the next state S_{t+1} and reward.

3.4. Evaluation Metrics

We evaluate the monitoring performance of the proposed RL-based EWMA chart using two popular SPC metrics, namely *ARL* for specific shift magnitudes, and an overall performance index based on Extra Quadratic Loss (EQL). The *ARL* is defined as the expected number of observations until a signal is triggered. Two variants are typically reported. ARL_0, the expected run length when the process is IC i.e., there is no shift. Higher values reflect better false alarm resistance. $ARL_1(\delta)$, the expected run length when a shift of magnitude δ occurs. The lower the ARL_1 and EQL values, the better the chart is in detecting shifts. The *ARL* values are individually computed for each shift magnitude δ in the test set. This provides information on the sensitivity of different magnitudes of the process changes. The EQL provides an aggregate measure of the efficiency of a chart across multiple shift magnitudes and is mathematically expressed as follows:

$$\text{EQL} = \int_{\delta_{\min}}^{\delta_{\max}} \delta^2\, \text{ARL}(\delta)\, d\delta \tag{9}$$

This formulation emphasizes the overall performance of the chart over a range of shifts rather than specific deviations (Wu et al., 2010). It is to be noted that the EQL integral is approximated numerically over the discrete shift grid using the trapezoidal rule.

3.5. Training Process of the Proposed Chart

The RL agent is trained once and then converted into a deterministic control chart using a probability threshold. Therefore, training and evaluation use different shift mechanisms but the same learned policy.

During training, the environment operates with the geometric shift mechanism described above. While $H_t = 0$, a Bernoulli trial with probability p determines whether a shift should be introduced. The shift magnitudes δ are sampled independently from a range $\delta \sim U[0.5, 3.0]$, so the agent sees a mixture of small and large shifts. The agent observes X_t and Z_t, selects $A_t \in \{0, 1\}$, and receives the corresponding reward. A neural network policy π_θ is trained using the PPO algorithm to maximize the expected discounted return. After training , we retain the policy that provides, for any

Table 2: Important Notations and Definitions used in this study

Symbol	Definition
X_t	Raw process observation at time t, $X_t \sim \mathcal{N}(\mu_t, \sigma^2)$
Z_t	EWMA statistic at time t: $Z_t = \lambda X_t + (1-\lambda)Z_{t-1}$
λ	Smoothing parameter for EWMA, $0 < \lambda < 1$
A_t	Agent action at time t: $A_t \in \{0$: wait, 1: signal$\}$
H_t	Hidden process state: $H_t = 0$ (IC), $H_t = 1$ (OOC)
p	Shift probability during training
τ	Shift time
R_t	Reward at time t, based on sampling, delay, and signaling accuracy
ARL_0	Average run length (ARL) under IC conditions
ARL_1	Average run length under OOC conditions
η	Decision threshold on $\pi_\theta(A_t = 1 \mid X_t, Z_t)$ for binary signal
π_θ	Stochastic policy parameterized by neural network weights
δ	Magnitude of process mean shift in units of σ
c_s	Cost of sampling when in control
c_d	Penalty for delayed detection: sampling while OOC
P_{FA}	Penalty for false alarms
R_{sig}	Reward for correct signaling when process is OOC

$X_t,\ Z_t$, a probability

$$p_t = \pi_\theta(A_t = 1 \mid X_t,\ Z_t)$$

interpreted as the estimate that signaling is optimal at t.

For performance evaluation, the environment is switched to a deterministic mode to match standard SPC practice. For a given shift magnitude δ and shift time τ:

- For IC experiments ($\delta = 0$), the process mean remains at μ_0 throughout the run.
- For OOC experiments ($\delta > 0$), the mean is μ_0 for $t < \tau$ and $\mu_0 + \delta\sigma$ for $t \geq \tau$.

This allows a direct comparison between the RL-based chart and the classical EWMA chart designed for the same λ and ARL_0.

The neural network policy is stochastic. To obtain a deterministic chart comparable to a classical EWMA scheme, the probability p_t is converted to a binary decision using a threshold $\eta \in [0, 1]$:

$$\delta_\eta(X_t, Z_t) = \begin{cases} 1, & \pi_\theta(A_t = 1 \mid X_t,\ Z_t) \geq \eta, \\ 0, & \pi_\theta(A_t = 1 \mid X_t,\ Z_t) < \eta, \end{cases} \tag{10}$$

where $\delta_\eta(X_t,\ Z_t) = 1$ denotes a signal. The IC and OOC $ARLs$ are:

$$\text{ARL}_0(\eta) = \mathbb{E}_0[T_\eta], \qquad \text{ARL}_1(\eta; \delta, \tau) = \mathbb{E}_{\delta,\tau}[T_\eta - \tau \,\big|\, T_\eta \geq \tau].$$

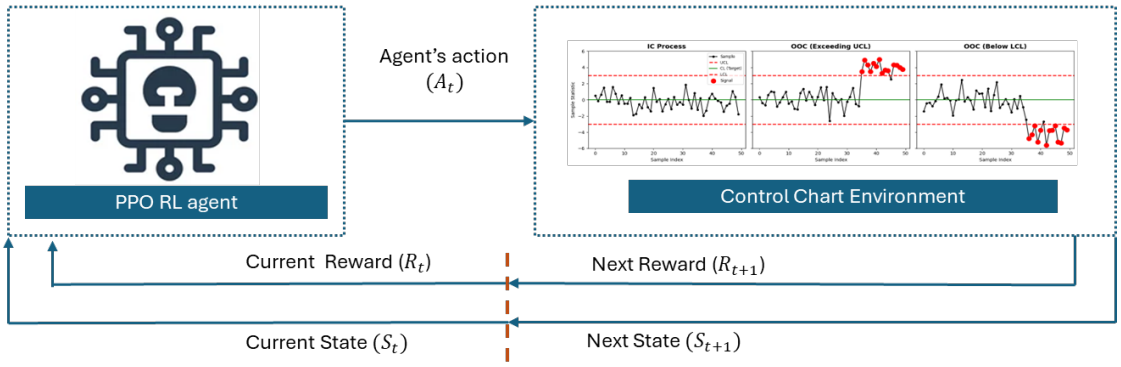


Figure 2: The proposed method interaction loop between the PPO agent and the control chart environment, showing state transitions ($S_t \rightarrow S_{t+1}$), rewards ($R_t \rightarrow R_{t+1}$), and the agent's action A_t.

To place the RL chart on the same footing as a classical EWMA, the threshold η is calibrated so that $\mathrm{ARL}_0(\eta)$ matches a prescribed design value $ARL_0^\star$ (for example, 500). Empirically, $ARL_0(\eta)$ is monotone increasing in η, so a simple bisection search is sufficient. Starting with $\eta_L = 0$ and $\eta_U = 1$, the trial values $\eta = (\eta_L + \eta_U)/2$ are tested by Monte Carlo simulation of IC trajectories, and the interval is tightened until $ARL_0(\eta)$ is within a tolerance of $ARL_0^\star$.

Once the calibrated value $\eta^\star$ is obtained, all the results of ARL_0 and ARL_1 reported for the RL chart are based on the deterministic rule (10) with $\eta = \eta^\star$.

For completeness, we summarize the RL training procedure and the *ARL* evaluation with threshold calibration. The algorithms are presented in Algorithms 2 and 3.

Algorithm 2: Agent training using PPO

Inputs : Smoothing parameter λ, reward parameters $(c_s, c_d, R_{\text{sig}}, P_{\text{FA}})$, PPO hyper-parameters;
Outputs : Trained policy π_θ;
Initialization: Policy network π_θ, value network V_ϕ, and optimizer;

Set shift probability p;
for *iter* $= 1, \ldots, N_{train}$ **do**
  Reset the environment to IC, set $Z_0 = 0$ and $H_0 = 0$;
  for $t = 1, \ldots, T_{rollout}$ **do**
    if $H_t = 0$ **then**
      Generate a shift with probability p;
      if *a shift occurs* **then**
        $H_t \leftarrow 1$ and fix the shifted mean $\mu_0 + \delta\sigma$ for the rest of the episode;
    Sample $X_t \sim \mathcal{N}(\mu(H_t), \sigma^2)$ and update $Z_t = \lambda X_t + (1-\lambda) Z_{t-1}$;
    Sample action A_t from the current policy $A_t \sim \pi_\theta(\cdot \mid X_t,\ Z_t)$;
    Compute reward R_t from the cost structure above;
    if $A_t = 1$ **then**
      Terminate the episode;
    Store $(X_t, Z_t, A_t, R_t, Z_{t+1})$ in a trajectory buffer;
  From the collected batch, compute advantage estimates and the PPO objective;
  Update θ and ϕ by several epochs of stochastic gradient steps on the PPO objectives.

4. Results and Discussion

In this section, we will present the numerical results of the study after stating the experimental setup. The comparative analysis with existing methods will also be discussed.

4.1. Experimental Setup

We simulate a Gaussian process with constant variance ($\sigma^2 = 1$) and mean shifts introduced at time points τ. Observations follow $X_t \sim \mathcal{N}(0,1)$ under the IC condition and $X_t \sim \mathcal{N}(\delta, 1)$ after the shift has occurred. We evaluate the chart in a wide range of shift magnitudes $\delta \in \{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 5.0, 6.0\}$ to capture both small and large shifts. We conducted the evaluation over 10,000 iterations per configuration reduce randomness in the results. Since the agent was trained on $\delta \in [0.5, 3.0]$ shifts, the values outside this range in the evaluation reflects generalization of the learned policy.

To account for the effect of the smoothing parameter in the EWMA statistic, we tested five smoothing parameters: $\lambda = 0.05, 0.14, 0.25, 0.5$, and 0.7. For each pair λ and δ, we generate multiple episodes to estimate ARL_0 and ARL_1. The shifts are introduced at $\tau = 1$, meaning that the ARL_1 results are zero-state.

Performance of the proposed chart is compared with CUSUM, classical and adaptive EWMA charts designed to achieve $\mathrm{ARL}_0 \approx 100$ for the first case and $\mathrm{ARL}_0 \approx 500$ in the second case. The *ARL* results presented in Tables 4 and 5 are replicated from Capizzi and Masarotto (2003) for the AEWMA, and computed using the R `spc` package (Knoth et al., 2016) for the EWMA and CUSUM charts. All the PPO hyper-parameters, environment settings, and reward values used to generate the proposed chart numerical results are stated in Table 8 in the Appendix.

4.2. Comparative Analysis

This section compares the proposed RL-based EWMA chart with AEWMA (Capizzi and Masarotto, 2003), classical EWMA, and CUSUM charts at matched IC behavior ($ARL_0 \approx 100$ and $ARL_0 \approx 500$). Since ARL_0 is calibrated to be approximately equal in all methods, the comparison focuses on (i) the behavior of ARL_1 in shift magnitudes and (ii) the aggregate EQL metric. We summarize the findings as follows:

Algorithm 3: Threshold calibration and ARL evaluation

Inputs : Trained policy π_θ, target $ARL_0^\star$, tolerance ε, maximum bisection iterations $K_{\max}$;

Outputs : Calibrated threshold $\eta^\star$, ARL_0 and ARL_1 estimates for the RL-EWMA chart;

Calibration of $\eta^\star$:;
Initialize $\eta_L = 0$, $\eta_U = 1$;
for $k = 1, \ldots, K_{\max}$ **do**
  Set $\eta_k = (\eta_L + \eta_U)/2$;
  Simulate many IC trajectories using the deterministic rule $\delta_{\eta_k}(X_t, Z_t)$ in (10); record the stopping times T_{η_k} and compute $\widehat{ARL}_0(\eta_k)$ as their sample mean;
  if $\left|\widehat{ARL}_0(\eta_k) - ARL_0^\star\right| \le \varepsilon$ **then**
    Set $\eta^\star = \eta_k$ and stop;
  if $\widehat{ARL}_0(\eta_k) < ARL_0^\star$ **then**
    Set $\eta_L = \eta_k$;
  else
    Set $\eta_U = \eta_k$;
ARL **evaluation with $\eta^\star$:**;
for *each shift magnitude δ and shift time τ of interest* **do**
  Simulate K trajectories with deterministic mean change at τ and charting rule $\delta_{\eta^\star}(X_t, Z_t)$;
  if $\delta = 0$ *(IC)* **then**
    Record all run lengths $T_{\eta^\star}$ and compute ARL_0 as their average;
  else
    Retain only runs with $T_{\eta^\star} \ge \tau$ and record detection delays $T_{\eta^\star} - \tau$; compute ARL_1 as their mean.

- For $ARL_0 \approx 100$ (Tables 3 & 5), the proposed RL-based EWMA generally produces smaller ARL_1 values than AEWMA, classical EWMA and CUSUM for small and moderate shifts ($\delta \in \{0.25, 0.50, 0.75, 1.00, 1.50\}$). For example, at $\delta = 0.50$, the ARL_1 values for RL-based chart range from 13.12 to 20.37 for different λ values, while the AEWMA values range from 17.64 to 20.30, the classical EWMA values range from 17.41 to 36.60, and the CUSUM values range from 19.33 to 48.83 depending on the reference parameter. At $\delta = 1.00$, RL-based ARL_1 values lie between 5.66 and 6.83, compared with AEWMA values between 6.92 and 9.38, classical EWMA values between 7.02 and 10.99, and CUSUM values between 7.39 and 16.39. For larger shifts ($\delta \ge 3.00$), all methods approach $ARL_1 \approx 1$, and the performance differences become small.

- Taking into account the overall performance, the proposed chart achieves lower EQL values (15.54–15.79) than AEWMA (17.66–18.94), classical EWMA (16.62–26.60), and CUSUM (16.68–26.18). This indicates improved overall detection performance of the proposed chart.

- For $ARL_0 \approx 500$ (Tables 3 & 4), similar trends are observed. For small shifts, the RL-based EWMA remains consistently faster than the competing charts. For instance, at $\delta = 0.25$, RL-based ARL_1 values range from 61.64 to 163.93, while AEWMA ranges from 98.51 to 119.45, classical EWMA from 84.01 to 308.89, and CUSUM from 94.80 to 362.50. At $\delta = 0.50$, RL-based ARL_1 ranges from 24.00 to 62.93, compared with AEWMA values between 33.70 and 40.94, classical EWMA values between 28.76 and 129.16, and CUSUM values between 31.08 and 183.68. For moderate shifts ($\delta \in [1.00, 2.50]$), the RL-based EWMA remains competitive and is often better than the other methods, while for larger shifts ($\delta \ge 4.00$) all charts again converge to $ARL_1 \approx 1$.

- In terms of EQL, the RL-based EWMA yields smaller values (18.85–20.94) than AEWMA (20.17–22.54), classical EWMA (21.03–34.29), and CUSUM (20.60–35.86). The minimum EQL among all the methods reported for $ARL_0 \approx 500$ is achieved by the RL-based EWMA with $\lambda = 0.50$ (EQL = 18.85), indicating the overall strongest performance under the tighter IC constraint.

Plots in Figure 3 visualize these comparisons using for the following parameters: RL-based EWMA with $\lambda = 0.50$, AEWMA with φ_{bs}, classical EWMA with $\lambda = 0.50$ and CUSUM with $k = 1.00$. In both figures, the *ARL* curve of the proposed RL-based EWMA lies below the competing methods for most shift magnitudes, particularly in the small to moderate shift region.

5. Additional Considerations and Extensions

The results presented thus far focus on the primary design and evaluation of the proposed RL-based

Table 3: RL-based EWMA ARL values across shifts δ and smoothing parameters λ with $ARL_0 \approx 100$ and $ARL_0 \approx 500$.

	$\text{ARL}_0 \approx 100$					$\text{ARL}_0 \approx 500$				
δ	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$
0.00	103.68	102.57	100.32	103.53	99.22	507.45	506.90	508.34	498.15	505.10
0.25	26.63	30.30	31.92	41.77	41.83	61.64	81.21	104.91	143.52	163.93
0.50	13.12	13.94	14.40	18.62	20.37	24.00	26.70	34.07	51.32	62.93
0.75	8.17	8.17	8.26	10.33	10.90	14.09	13.78	15.54	22.53	27.76
1.00	5.94	5.76	5.66	6.43	6.83	9.79	8.82	9.44	11.95	14.53
1.50	3.62	3.45	3.33	3.41	3.48	6.00	5.07	4.81	5.08	5.65
2.00	2.50	2.40	2.38	2.29	2.27	4.21	3.55	3.23	3.10	3.22
2.50	1.86	1.82	1.87	1.73	1.72	3.15	2.73	2.46	2.26	2.23
3.00	1.45	1.47	1.53	1.41	1.40	2.43	2.18	1.99	1.80	1.79
3.50	1.23	1.23	1.30	1.21	1.21	1.92	1.82	1.67	1.51	1.51
4.00	1.09	1.10	1.14	1.10	1.09	1.54	1.54	1.44	1.40	1.29
5.00	1.01	1.01	1.02	1.01	1.01	1.13	1.16	1.12	1.06	1.06
6.00	1.00	1.00	1.00	1.00	1.00	1.02	1.02	1.01	1.01	1.00
EQL	15.59	15.55	15.79	**15.54**	15.57	20.94	20.12	19.28	**18.85**	19.41

EWMA chart under the assumption that process observations are normally distributed with known parameters, and that any shift occurs immediately at the start (zero-state). In this section, we examine and discuss some additional considerations that are important for practical implementation. These include the effect of parameter estimation, the role of the normality assumption, and steady-state *ARL* performance.

5.1. Effect of Parameter Estimation

Standard control chart designs usually assume that the IC process parameters, i.e., the mean μ_0 and standard deviation σ_0, are known, also called case K. Under this assumption, the chart parameters are calibrated to achieve a specified IC ARL_0. In practice, μ_0 and σ_0 are rarely known and must therefore be estimated from a Phase I reference sample; this is known as case U. However, replacing (μ_0, σ_0) with estimates $(\hat{\mu}_0, \hat{\sigma}_0)$ introduces additional randomness into the Phase II monitoring and affects both the IC and OOC run length behavior (Jensen et al., 2006; Castagliola et al., 2016). For EWMA charts, Jones et al. (2001) provides a detailed analysis of the effect of parameter estimation.

To account for parameter estimation, a two-phase procedure is usually adopted. In Phase I, an IC reference sample is collected and used to estimate μ_0 and σ_0. In Phase II, the control chart is applied using these estimates, and the run length is recorded conditional on the outcome of Phase I. Jones et al. (2001) considers several unbiased estimators of σ_0, including $\bar{R}/d_2$, $\bar{S}/c_4$, and the pooled estimator $S_p/c_{4,m}$, and adopts the latter due to its efficiency. In this study, we focus on the individual observations case, which represents the analog of their method when the subgroup size is equal to one. This approach was also adopted by Atalay et al. (2020). For a Phase I reference sample of size m, we estimate

$$\hat{\mu}_0 = \frac{1}{m}\sum_{i=1}^{m} X_i,$$

$$\hat{\sigma}_0 = \frac{s}{c_4(m-1)},$$

where s is the sample standard deviation and $c_4(\cdot)$ is the correction factor commonly tabulated in the SPC references (see Montgomery (2020)).

The RL approach adopted in this study is in parallel with the classical design used in traditional EWMA charts. Here, the PPO agent with the policy $\pi_\theta(\cdot)$ is trained under a known case K condition, where the process follows $N(0, 1)$, exactly as discussed in Section 3.2. Then a decision threshold is calibrated to achieve a specified ARL_0 in this known environment, playing a role analogous to the selection of control limits in the standard EWMA design. To assess the effect of parameter estimation, we applied the calibrated threshold without modification to monitor the process in the unknown case. In this case, each evaluation episode begins with a new Phase I reference sample, from which $(\hat{\mu}_0, \hat{\sigma}_0)$ are calculated and used to standardize the Phase II observations.

Tables 6 and 7 summarize the resulting run length distributions in terms of *ARL*, *SDRL*, $Q_{0.10}$, median, and

Table 4: ARL values for AEWMA, EWMA, and CUSUM charts with $\text{ARL}_0 \approx 500$.

	AEWMA			EWMA					CUSUM				
δ	φ_{hu}	φ_{bs}	φ_{cb}	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	$k = 0.25$	$k = 0.50$	$k = 0.75$	$k = 1.00$	$k = 2.00$
0.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00
0.25	98.51	105.87	119.45	84.01	124.48	170.34	254.83	308.89	94.80	145.53	200.53	249.52	362.50
0.50	40.94	33.70	33.76	28.76	34.96	48.30	88.81	129.16	31.08	38.87	57.04	81.44	183.68
0.75	25.04	17.95	16.16	16.37	16.44	20.12	35.92	56.20	17.54	17.32	22.12	30.90	87.88
1.00	17.59	11.78	10.16	11.38	10.21	11.14	17.48	27.29	12.17	10.52	11.60	14.67	43.50
1.50	10.11	6.50	5.77	7.11	5.73	5.46	6.53	8.97	7.58	5.82	5.45	5.75	12.90
2.00	6.08	4.16	4.06	5.22	4.03	3.61	3.63	4.26	5.55	4.06	3.55	3.41	5.21
2.50	3.66	2.89	3.15	4.17	3.15	2.74	2.50	2.61	4.41	3.15	2.67	2.45	2.81
3.00	2.29	2.14	2.56	3.50	2.61	2.26	1.93	1.86	3.69	2.60	2.18	1.94	1.87
3.50	1.60	1.67	2.10	3.04	2.26	1.95	1.58	1.47	3.19	2.25	1.87	1.62	1.43
4.00	1.26	1.37	1.75	2.69	2.04	1.73	1.34	1.24	2.84	2.03	1.63	1.38	1.20
5.00	1.04	1.08	1.27	2.16	1.75	1.32	1.07	1.04	2.26	1.72	1.24	1.09	1.03
6.00	1.00	1.01	1.05	2.00	1.37	1.07	1.01	1.00	2.02	1.33	1.04	1.01	1.00
EQL	22.19	20.17	22.54	34.29	26.46	22.33	21.03	22.89	35.86	26.30	21.69	20.60	26.44

Table 5: ARL values for AEWMA, EWMA, and CUSUM charts with $\text{ARL}_0 \approx 100$.

	AEWMA			EWMA					CUSUM				
δ	φ_{hu}	φ_{bs}	φ_{cb}	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	$k = 0.25$	$k = 0.50$	$k = 0.75$	$k = 1.00$	$k = 2.00$
0.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
0.25	41.38	40.53	46.53	38.80	44.13	50.77	63.15	71.08	44.54	52.66	60.86	67.86	80.21
0.50	20.30	17.64	18.96	17.41	18.18	20.84	28.91	36.60	19.33	21.70	26.60	32.55	48.83
0.75	13.01	10.59	10.41	10.82	10.30	11.02	14.66	19.15	11.57	11.46	13.28	16.25	27.94
1.00	9.38	7.43	6.92	7.83	7.02	7.06	8.60	10.99	8.19	7.39	7.90	9.21	16.39
1.50	5.67	4.52	4.10	5.09	4.28	3.99	4.14	4.77	5.19	4.25	4.04	4.21	6.57
2.00	3.68	3.15	2.95	3.82	3.12	2.81	2.62	2.74	3.84	3.01	2.70	2.61	3.30
2.50	2.46	2.37	2.35	3.09	2.50	2.21	1.93	1.89	3.08	2.38	2.05	1.90	2.03
3.00	1.73	1.87	1.99	2.61	2.13	1.85	1.54	1.46	2.59	2.00	1.67	1.52	1.47
3.50	1.34	1.53	1.74	2.28	1.88	1.58	1.30	1.23	2.26	1.73	1.41	1.28	1.21
4.00	1.14	1.30	1.52	2.08	1.68	1.37	1.14	1.10	2.06	1.51	1.22	1.13	1.08
5.00	1.02	1.06	1.17	1.85	1.28	1.09	1.02	1.01	1.80	1.16	1.04	1.02	1.01
6.00	1.00	1.01	1.02	1.51	1.06	1.01	1.00	1.00	1.44	1.02	1.00	1.00	1.00
EQL	17.66	17.84	18.94	26.60	20.30	17.87	16.62	16.86	26.18	19.04	17.06	16.68	18.14

$Q_{0.90}$ across δ values. Reporting dispersion and quantiles alongside *ARL* is important because Phase I estimation produces a mixture of conditional run length distributions, and the mean can be strongly influenced by long runs. A key empirical feature of these results is that the IC ARL_0 in case U departs from its nominal design target, and that departure is most pronounced for small m. In addition, the IC run length distribution becomes more dispersed in case U, as reflected by the increase in $SDRL$ and the widening gap between $Q_{0.10}$ and $Q_{0.90}$.

The direction of the shift of the IC ARL_0, either inflation or deflation relative to the nominal target, is not universal in the classical literature. For subgroup EWMA charts, Jones et al. (2001) showed that parameter estimation can produce substantially more frequent false alarms on average while also degrading the sensitivity of OOC. For individual EWMA charts, Human et al. (2011) stated that the performance of case U can deviate from case K in either direction depending on the

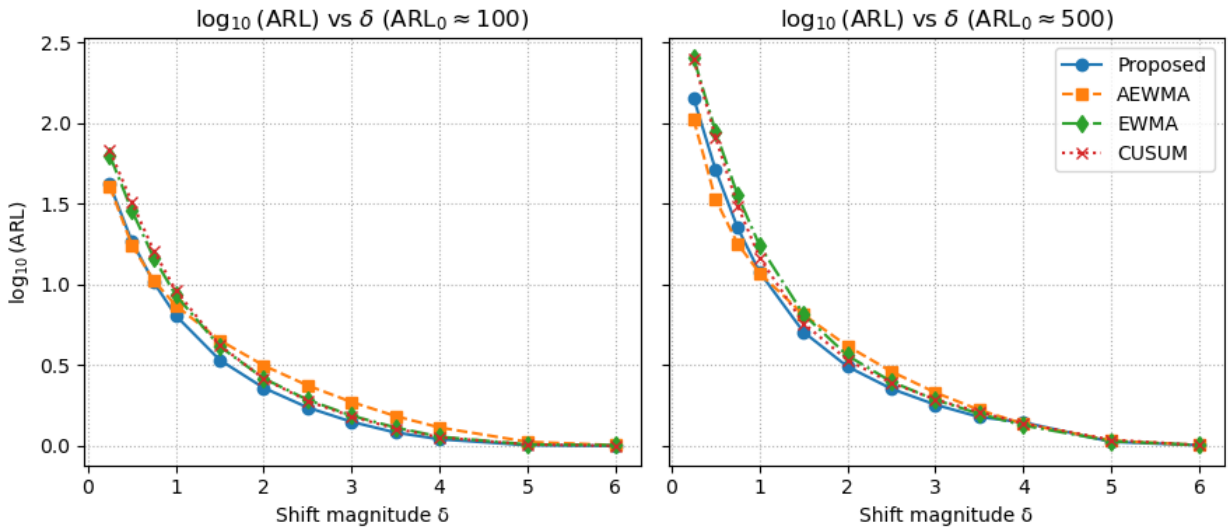


Figure 3: $\log_{10}(ARL)$ versus shift magnitude δ for $ARL_0 \approx 500$ comparing the proposed RL-EWMA with AEWMA, classical EWMA, and CUSUM charts.

estimator and the design and that *ARL* alone can mask substantial distributional changes. In our proposed design, the calibrated $\eta^\star$ is fixed from case K and is not re-tuned under estimation, so any IC ARL_0 inflation observed in case U can be interpreted as the policy becoming effectively more conservative under the altered state distribution induced by $(\hat{\mu}_0, \hat{\sigma}_0)$. Practically, this effect diminishes as m increases, consistent with the classical expectation that the run length distribution under estimated parameters approaches the case K behavior as the Phase I reference sample is increased.

Across both targets, Tables 7 and 6 show that OOC $ARL_1(\delta)$ in case U is typically larger than in case K (Table 3) for the same δ, which is consistent with the case U EWMA literature (Jones et al., 2001; Human et al., 2011). From the RL perspective, this corresponds to evaluating the decision rule induced by $(\pi_\theta, \eta^\star)$ on a different state distribution: in case U the standardized shift uses episode-specific $(\hat{\mu}_0, \hat{\sigma}_0)$, so the realized EWMA state s_t is subject to random location and scale distortions even when the underlying data remain normal. Keeping $\eta^\star$ fixed therefore changes the frequency with which trajectories enter regions where $\pi_\theta(1 \mid s_t) > \eta^\star$, which shifts both the false alarm behavior and OOC detection delay. From Tables 6 and 7, we observe the following:

- IC ARL_0 in case U increases sharply for small m, and the effect strengthens as the ARL_0 target increases. For the $ARL_0 \approx 100$ design with $\lambda = 0.05$, the mean of IC of case U is 991.98 at $m = 20$, and drops to 350.11 at $m = 50$ and 184.36 at $m = 100$ (Table 6). A similar pattern is observed for other values of λ, e.g., for $\lambda = 0.14$ the case U IC mean is 537.25 ($m = 20$), 204.15 ($m = 50$), and 140.84 ($m = 100$). This is the case at $ARL_0 \approx 500$. For $\lambda = 0.05$, the case U IC mean is 2665.26 ($m = 20$), 1788.99 ($m = 50$), and 1254.55 ($m = 100$), while for $\lambda = 0.75$ it is 1411.66 ($m = 20$), 886.33 ($m = 50$), and 696.75 ($m = 100$) (Table 7). This effect of IC ARL_0 increasing in case U contrasts with the more commonly reported tendency toward increased IC false alarms for some classical EWMA designs (Jones et al., 2001; Human et al., 2011).

- The OOC performance still deteriorates in case U, but the magnitude depends on δ and is most visible for small to moderate shifts. In the $ARL_0 \approx 100$ design with $\lambda = 0.05$, $ARL_1(0.50)$ is 16.70 at $m = 20$ and decreases to 13.42 at $m = 50$ and 12.67 at $m = 100$; meanwhile $ARL_1(1.00)$ is 6.06 ($m = 20$), 5.86 ($m = 50$), and 5.84 ($m = 100$) (Table 6). For larger shifts the mean quickly approaches 1–2 across all m (e.g., at $\delta = 6.00$, means are 1.11, 1.08, and 1.07 for $m = 20, 50, 100$ when $\lambda = 0.05$), so the practical differences concentrate in the small to moderate shift regime that is also emphasized in classical EWMA comparisons.

- The upper-tail behavior becomes more consequential when m is small and when the target ARL_0 is large. For $ARL_0 \approx 100$ and $\lambda = 0.05$, the IC $Q90$

is 2792.50 at $m = 20$ versus 661.10 at $m = 50$ and 397.00 at $m = 100$; for $ARL_0 \approx 500$ and $\lambda = 0.05$, the IC $Q90$ reaches 10000.00 at $m = 20$ and remains large even at $m = 50$ (7993.60) and $m = 100$ (3500.50).

- The dependence on λ is present but secondary to the Phase I sample size m. At $ARL_0 \approx 100$ and $m = 20$, ARL_0 decreases as λ increases from 0.05 to 0.75 (from 991.98 to 280.07), but it remains far above the target in all cases (Table 3). At $m = 100$, the IC means are closer together in λ (e.g., 184.36 at $\lambda = 0.05$ and 117.59 at $\lambda = 0.75$).

These results can be interpreted as a reuse effect in the sense that in classical EWMA, we fix (λ, L) under known parameters and then evaluate the *ARL* performance when these parameters are estimated, while the RL design fixes $\eta^\star$ that is calibrated in case K and reuses it in case U. Conistent with the literature, we see that a large reference sample is need to achieve similar performance as the known case. Figure 4 presents some visual comparisons of the ARL profiles of the proposed chart under the known and unknown parameter cases.

5.2. Conditional Expected Delay and Steady-State Performance

The results presented thus far are based on the zero-state *ARL*, which assumes that monitoring begins at chart initialization ($\tau = 1$) and that any process change occurs immediately. While this measure is convenient for design and comparison, it does not fully capture detection behavior in realistic monitoring environments. For this reason, alternative performance measures like conditional expected delay (CED) and the steady-state average run length are usually reported alongside the zero-state results.

The CED is the expected number of observations required to detect a change after it has occurred on condition that the process has been under IC prior to the change and that no false alarms were raised. Formally, the CED can be expressed as

$$\mathrm{CED}_\tau = \mathbb{E}\big[(\mathit{Run\ Length} - \tau + 1) \mid \mathit{Run\ Length} \geq \tau\big].$$

The steady-state *ARL* provides a complementary perspective on long-term performance. It is defined as

$$ARL_{\mathrm{ss}} = \lim_{\tau \to \infty} \mathbb{E}\big[(\mathit{Run\ Length} - \tau + 1) \mid \mathit{Run\ Length} \geq \tau\big],$$

and can be interpreted as the limiting value of CED as the change-point $\tau \to \infty$. The steady-state *ARL* is widely regarded as a more realistic measure of chart performance in continuous monitoring environments, where a process may operate in control for an extended period before a change occurs. Ibrahim et al. (2025) present a detailed discussion and results on CED and steady-state *ARL* performance of EWMA, CUSUM, and Mixed EWMA-CUSUM charts.

For the purpose of this comparison, all charts are designed to achieve a target of $ARL_0 \approx 100$. The smoothing parameter for both the proposed RL-based chart and the classical EWMA chart is set to $\lambda = 0.05$, and the reference parameter for the CUSUM chart is $k = 0.25$. The other designs we investigated led to similar conclusions and are therefore omitted here for brevity.

The CED profiles illustrated in Figure 5 reveal a distinct detection performance between the RL approach and traditional methods. Some important observations are as fpllows:

- zero-state vs. steady-state behavior: A unique characteristic of the proposed RL-based chart is that its CED values increase from the initial state ($\tau = 1$) up to a specific change point before stabilizing into a steady-state. In contrast, both the classical EWMA and CUSUM charts exhibit CED values that decrease as the change point τ increases.

- At $\delta = 0.25$: The performance advantage of the RL-based chart is most pronounced for small shifts. It stabilizes at a steady-state CED between approximately 25 and 28, whereas the classical EWMA and CUSUM charts level off at significantly higher values of about 34 and 36, respectively.

- At $\delta = 0.5$, the steady-state CED based on RL is approximately 14, outperforming CUSUM (≈ 15) and EWMA (≈ 16).

- At $\delta = 1.0$, the RL-based chart maintains a steady-state delay of approximately 6.7, compared to 7.4 for the classical EWMA.

- At $\delta = 2.0$: Even for larger shifts, the RL-based lead persists with a steady-state CED of approximately 2.9, while CUSUM and EWMA stabilize at 3.0 and 3.75, respectively.

We can draw from these results that the signaling policy learned by the PPO agent is more efficient for continuous process monitoring than the classical control charts. The proposed chart achieves superior steady-state performance across a wide range of shift magnitudes, in addition to the superior performance presented in section 4

Table 6: RL-based EWMA chart performance (Case U) for three reference sample sizes $m \in \{20, 50, 100\}$ and smoothing parameter $\lambda \in \{0.05, 0.14, 0.50, 0.75\}$. For each shift size δ, we report ARL, SDRL, Q10, Median, and Q90. The charts are designed to achieve an $ARL_0 \approx 100$ under case K.

		$m = 20$				$m = 50$				$m = 100$			
δ	Stat	$\lambda = 0.05$	0.14	0.50	0.75	$\lambda = 0.05$	0.14	0.50	0.75	$\lambda = 0.05$	0.14	0.50	0.75
0.00	ARL	991.98	537.25	294.99	280.07	350.11	204.15	151.50	142.78	184.36	140.84	124.53	117.59
	SDRL	2505.22	1608.26	874.79	820.23	1100.18	552.85	277.04	236.02	461.16	243.57	179.81	157.87
	Q10	9.00	9.00	8.00	8.00	11.00	10.00	9.00	9.00	12.00	11.00	11.00	10.00
	Median	63.00	66.00	66.00	68.00	65.00	68.00	67.00	68.00	68.50	70.00	68.00	67.00
	Q90	2792.50	1012.00	603.00	601.10	661.10	444.00	359.10	335.00	397.00	322.00	293.00	278.00
0.25	ARL	79.65	71.34	87.08	102.25	34.96	39.17	53.31	61.22	28.05	32.75	45.37	52.91
	SDRL	340.07	223.02	224.92	273.52	70.26	64.58	86.43	93.20	31.14	37.34	55.89	63.62
	Q10	6.00	5.00	4.00	4.00	6.00	6.00	5.00	5.00	6.00	6.00	5.00	5.00
	Median	18.00	21.00	27.00	31.00	18.00	20.00	27.00	32.00	18.00	21.00	27.00	32.00
	Q90	121.00	139.00	194.00	218.00	70.00	85.00	122.00	144.00	59.00	73.00	106.00	125.00
0.50	ARL	16.70	20.10	32.38	42.46	13.42	14.66	23.27	29.24	12.67	13.83	20.15	26.23
	SDRL	29.99	39.87	76.80	98.15	12.01	14.59	33.64	39.97	9.74	11.84	21.84	30.44
	Q10	4.00	4.00	3.00	3.00	5.00	4.00	3.00	3.00	5.00	4.00	3.00	3.00
	Median	10.00	10.00	13.00	16.00	10.00	10.00	13.00	16.00	10.00	10.00	13.00	16.00
	Q90	31.00	40.00	70.00	95.00	26.00	30.00	52.00	68.00	23.10	28.00	45.00	61.00
0.75	ARL	8.88	9.46	14.75	20.11	8.10	8.38	11.89	15.52	7.95	8.07	10.79	13.97
	SDRL	8.28	10.57	26.98	38.71	5.12	6.19	13.67	20.20	4.55	5.45	10.34	14.89
	Q10	3.00	3.00	2.00	2.00	3.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00
	Median	7.00	7.00	7.00	9.00	7.00	7.00	8.00	9.00	7.00	7.00	8.00	9.00
	Q90	16.00	18.00	32.00	46.00	14.00	16.00	26.00	35.00	14.00	15.00	23.00	31.00
1.00	ARL	6.06	6.20	8.10	11.58	5.86	5.83	6.99	9.22	5.84	5.66	6.65	8.65
	SDRL	3.70	4.82	11.75	22.69	2.94	3.53	6.57	10.52	2.81	3.19	5.52	8.63
	Q10	3.00	2.00	2.00	1.00	3.00	3.00	2.00	2.00	3.00	3.00	2.00	2.00
	Median	5.00	5.00	5.00	6.00	5.00	5.00	5.00	6.00	5.00	5.00	5.00	6.00
	Q90	10.00	11.00	17.00	25.00	10.00	10.00	14.00	20.00	9.00	10.00	14.00	19.00
1.50	ARL	3.97	3.71	3.77	4.58	3.88	3.60	3.52	4.06	3.87	3.60	3.41	3.95
	SDRL	1.76	1.91	3.41	5.32	1.52	1.64	2.47	3.67	1.46	1.55	2.21	3.21
	Q10	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Median	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00
	Q90	6.00	6.00	7.00	10.00	6.00	6.00	7.00	8.00	6.00	6.00	6.00	8.00
2.00	ARL	2.97	2.71	2.36	2.57	2.95	2.65	2.29	2.40	2.94	2.66	2.25	2.37
	SDRL	1.08	1.10	1.52	2.23	0.96	0.98	1.28	1.71	0.93	0.96	1.19	1.61
	Q10	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Median	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00
	Q90	4.00	4.00	4.00	5.00	4.00	4.00	4.00	4.10	4.00	4.00	4.00	4.00
2.50	ARL	2.42	2.17	1.73	1.75	2.42	2.16	1.71	1.69	2.41	2.17	1.71	1.65
	SDRL	0.77	0.78	0.88	1.13	0.69	0.69	0.81	0.96	0.66	0.67	0.80	0.90
	Q10	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Median	2.00	2.00	2.00	1.00	2.00	2.00	2.00	1.00	2.00	2.00	2.00	1.00
	Q90	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
3.00	ARL	2.08	1.85	1.43	1.37	2.07	1.85	1.40	1.34	2.08	1.83	1.39	1.33
	SDRL	0.60	0.62	0.62	0.68	0.53	0.58	0.58	0.62	0.51	0.56	0.57	0.58
	Q10	1.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Median	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Q90	3.00	3.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00
4.00	ARL	1.65	1.42	1.11	1.08	1.67	1.39	1.09	1.06	1.66	1.39	1.09	1.06
	SDRL	0.51	0.51	0.32	0.29	0.50	0.50	0.29	0.24	0.49	0.49	0.29	0.24
	Q10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	Median	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Q90	2.00	2.00	2.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00

Table 7: RL-based EWMA chart performance (Case U) for three reference sample sizes $m \in \{20, 50, 100\}$ and smoothing parameter $\lambda \in \{0.05, 0.14, 0.50, 0.75\}$. For each shift size δ, we report ARL, SDRL, Q10, Median, and Q90. The charts are designed to achieve an $ARL_0 \approx 500$ under case K.

		$m = 20$				$m = 50$				$m = 100$			
δ	Stat	$\lambda = 0.05$	0.14	0.50	0.75	$\lambda = 0.05$	0.14	0.50	0.75	$\lambda = 0.05$	0.14	0.50	0.75
0.00	ARL	2665.26	2058.48	1630.13	1411.66	1788.99	1284.27	973.51	886.33	1254.55	884.84	699.85	696.75
	SDRL	3923.61	3377.29	2866.42	2571.83	3040.54	2350.32	1782.43	1593.07	2276.50	1570.34	1104.48	1078.84
	Q10	24.00	22.00	24.00	25.00	36.00	33.00	35.00	36.00	46.00	43.00	41.00	44.00
	Median	322.50	312.00	323.00	310.00	331.50	323.00	320.00	315.00	343.00	343.00	330.00	336.00
	Q90	10000.00	10000.00	6316.40	4583.30	7993.60	3833.20	2490.10	2216.40	3500.50	2159.00	1711.10	1689.20
0.25	ARL	500.65	462.94	596.54	671.33	122.67	154.84	271.00	331.70	82.07	104.68	193.27	252.25
	SDRL	1702.48	1482.57	1598.13	1673.84	366.06	402.36	599.84	693.46	133.28	164.84	295.87	380.10
	Q10	12.00	10.00	10.00	11.00	14.00	12.00	14.00	15.00	15.00	13.00	15.00	17.00
	Median	46.00	55.00	97.00	121.00	46.00	56.00	99.00	122.00	46.00	55.00	99.00	128.00
	Q90	671.30	801.10	1245.50	1537.10	232.00	326.00	619.10	790.10	169.00	238.00	460.00	607.10
0.50	ARL	44.32	64.63	165.51	248.00	27.99	34.85	78.73	128.81	25.04	30.00	66.48	101.39
	SDRL	121.84	210.72	559.48	810.64	33.96	51.59	145.35	251.74	19.21	34.65	94.33	139.70
	Q10	8.00	6.00	5.00	6.00	9.00	7.00	6.00	7.00	9.00	7.00	7.00	9.00
	Median	20.00	20.00	34.00	51.00	20.00	20.00	37.00	54.00	20.00	20.00	38.00	57.00
	Q90	75.00	119.00	331.00	478.10	52.00	73.00	180.00	298.00	45.00	61.00	152.00	236.00
0.75	ARL	17.26	20.04	53.53	103.55	14.69	15.37	32.41	55.51	14.26	14.25	27.29	45.70
	SDRL	23.43	37.74	149.57	316.83	9.06	14.52	52.71	93.50	7.50	10.54	31.70	61.42
	Q10	6.00	5.00	3.00	4.00	7.00	5.00	4.00	4.00	7.00	5.00	4.00	5.00
	Median	12.00	11.00	17.00	24.00	13.00	11.00	17.00	27.00	13.00	11.00	17.00	26.00
	Q90	30.00	38.00	110.10	214.00	25.00	29.00	74.00	129.00	24.00	27.00	62.00	107.00
1.00	ARL	10.65	10.63	22.00	43.46	9.98	9.31	15.14	26.95	9.97	8.92	13.93	22.92
	SDRL	6.60	10.51	49.22	113.56	4.62	6.17	18.22	42.92	4.27	5.01	14.36	27.53
	Q10	5.00	4.00	3.00	3.00	5.00	4.00	3.00	3.00	6.00	4.00	3.00	3.00
	Median	9.00	8.00	9.00	14.00	9.00	8.00	9.00	14.00	9.00	8.00	9.00	14.00
	Q90	18.00	20.00	44.00	94.10	16.00	16.00	33.00	61.00	16.00	15.00	30.00	53.00
1.50	ARL	6.44	5.43	6.89	11.86	6.29	5.19	5.71	8.63	6.25	5.15	5.49	7.84
	SDRL	2.68	2.99	9.62	25.38	2.14	2.32	4.73	10.54	1.99	2.16	4.14	7.56
	Q10	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00
	Median	6.00	5.00	4.00	5.00	6.00	5.00	4.00	5.00	6.00	5.00	4.00	5.00
	Q90	10.00	9.00	14.00	25.00	9.00	8.00	11.00	19.00	9.00	8.00	10.00	17.00
2.00	ARL	4.67	3.73	3.50	4.70	4.65	3.67	3.29	3.96	4.61	3.68	3.23	3.84
	SDRL	1.52	1.51	2.86	6.46	1.34	1.30	2.11	3.75	1.24	1.24	1.86	3.13
	Q10	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00
	Median	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00
	Q90	7.00	6.00	6.00	10.00	6.00	5.00	6.00	8.00	6.00	5.00	6.00	8.00
2.50	ARL	3.73	2.92	2.37	2.67	3.71	2.91	2.28	2.41	3.70	2.89	2.26	2.36
	SDRL	1.09	1.00	1.42	2.73	0.93	0.88	1.15	1.67	0.89	0.85	1.06	1.54
	Q10	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00
	Median	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00
	Q90	5.00	4.00	4.00	5.00	5.00	4.00	4.00	4.00	5.00	4.00	4.00	4.00
3.00	ARL	3.13	2.44	1.80	1.83	3.12	2.42	1.78	1.71	3.13	2.40	1.77	1.69
	SDRL	0.83	0.74	0.86	1.21	0.73	0.65	0.78	0.96	0.70	0.62	0.74	0.90
	Q10	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Median	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00
	Q90	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00
4.00	ARL	2.41	1.89	1.29	1.21	2.40	1.91	1.26	1.18	2.38	1.91	1.25	1.17
	SDRL	0.57	0.52	0.49	0.47	0.53	0.46	0.46	0.41	0.50	0.43	0.45	0.41
	Q10	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Median	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Q90	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00

5.3. Normality Assumption and Possible extension to Non-normal Processes

In this study, both the training and evaluation stages assume a Gaussian IC process. This choice follows the classical literature on control charts. Maintaining this assumption allows the effect of Phase I parameter estimation to be examined in isolation, without confounding distributional misspecification. The robustness of

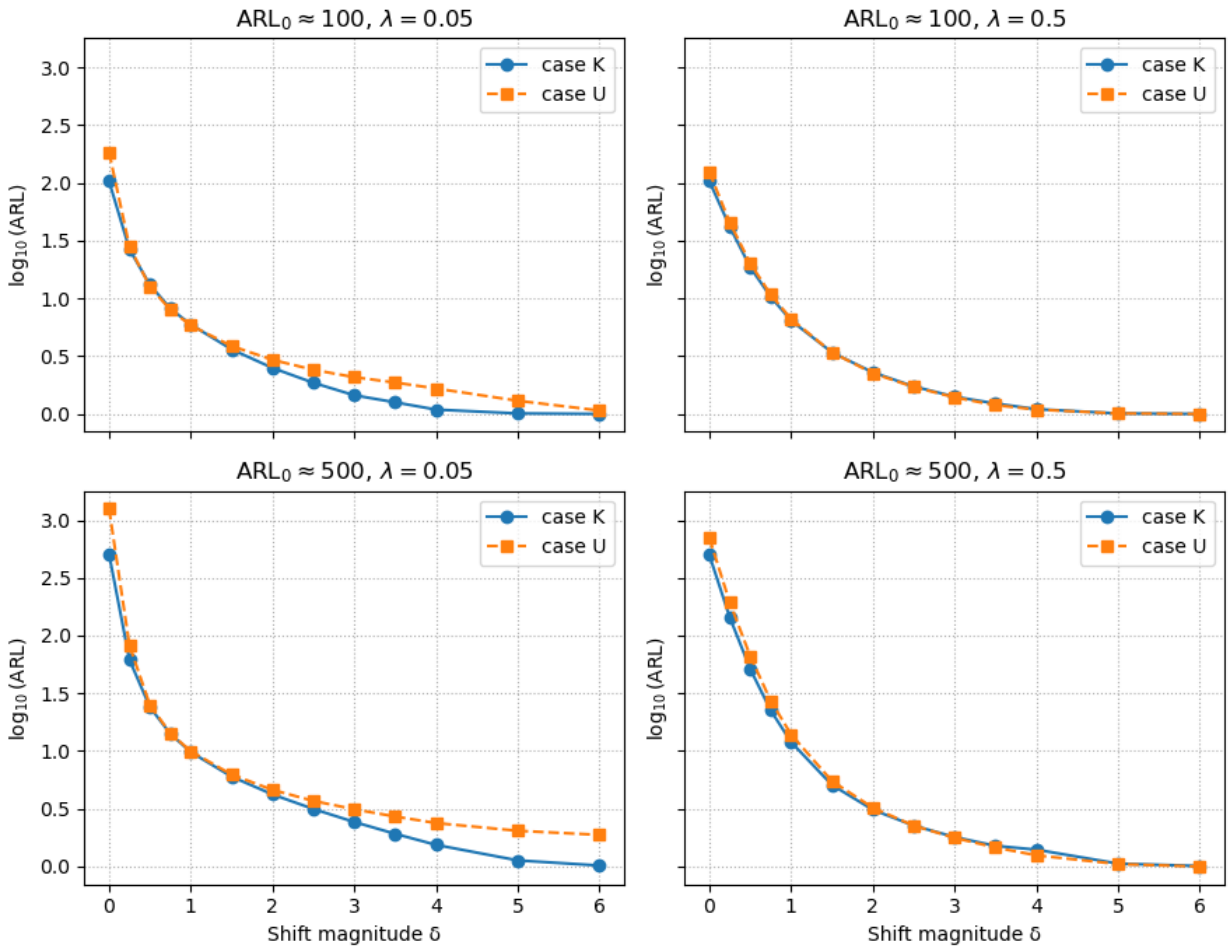


Figure 4: ARL comparison for known and unknown cases for selected values of λ (0.05, 0.5) and $m = 100$

EWMA charts to non-normal distributions has been investigated extensively. For example, Human et al. (2011); Borror et al. (1999), showed that EWMA charts with small smoothing constants can exhibit reasonable robustness to symmetric heavy tailed and mildly skewed distributions when parameters are known. This robustness is one reason why EWMA charts are often preferred over Shewhart-type charts in non-normal settings.

Considering the RL-based framework adopted here, non-normality represents an additional form of distribution shift beyond Phase I parameter estimation. The policy π_θ is trained under a Gaussian IC state distribution, and deviations from normality may alter how frequently the EWMA state visits regions associated with signaling. When combined with estimated parameters, this can substantially change both the IC false alarm behavior and the OOC detection delay. The current work focuses on the Gaussian process to isolate the effect of Phase I parameter estimation under reuse of a fixed decision threshold $\eta^\star$. Although we have conducted some preliminary robustness experiments under non-Gaussian distributions, a systematic analysis of non-normality, especially its interaction with parameter estimation and RL-based decision rules, is deferred to future work.

6. Case Study: Monitoring CTA Bus Ridership

6.1. The Data

In this section, we present a case study based on monthly bus ridership data from the Chicago Transit Authority (CTA). The data is obtained from the Chicago City Open Data Portal City of Chicago, Transportation (2026). The data set contains monthly total ridership counts and monthly day-type averages for each

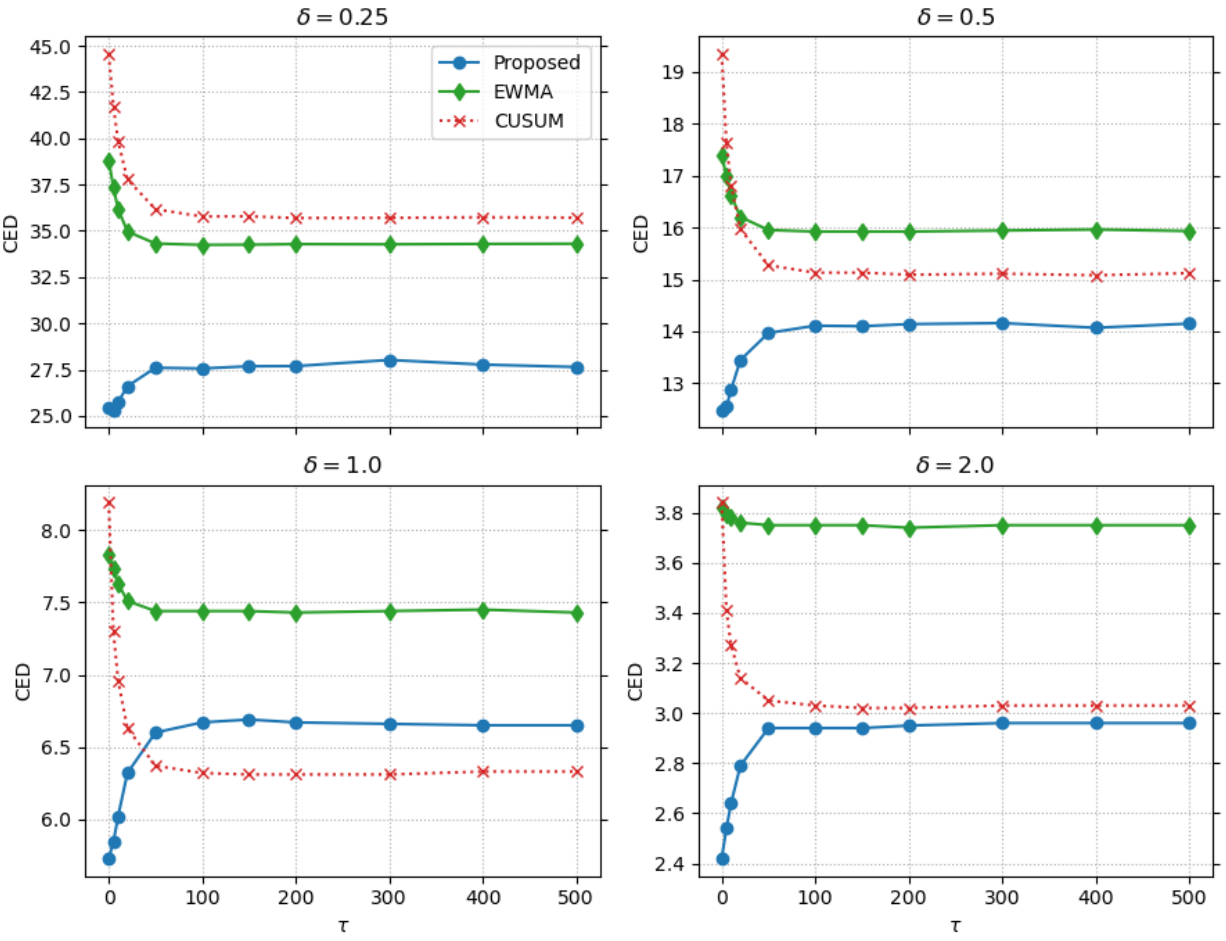


Figure 5: Conditional Expected Delay versus change point τ for the proposed RL-based, classical EWMA, and CUSUM charts across various shift magnitudes δ.

bus route. The series covers more than two decades, starting from January 2001 to November 2025. This includes periods of relative stability as well as multiple system-wide disruptions.

In this study, we focus on the total ridership. This choice yields a univariate series that is directly interpretable, meaningful, and commonly used in transit demand analysis Boisjoly et al. (2018). The CTA bus ridership data has some properties that are particularly important for monitoring applications. The series contains well-documented disruptions, including the 2008–2009 financial crisis and the COVID-19 pandemic, which provide natural stress tests for monitoring behavior. As a result, the data set allows us to assess whether the monitoring methods produce interpretable signals under real conditions rather than just under ideal assumptions.

6.2. Monitoring scenarios

To investigate different use cases, we consider two monitoring scenarios. Scenario A focuses on a single route, while Scenario B is concerned with a system-wide monitoring of all routes data available. All the charts are calibrated to achieve a comparable IC performance, with a target $\text{ARL}_0 \approx 500$. For the EWMA chart, the smoothing parameter is set to $\lambda = 0.50$, and the control limit factor is $L = 3.0701$. The CUSUM chart uses a reference value of $k = 0.5$ and a decision interval of $h = 5.0707$. The proposed RL-based chart also uses $\lambda = 0.50$ and converts the policy output into a signal using a probability threshold. This threshold is calibrated to $\eta = 0.25$ for the route-level and to $\eta = 0.50$ for the aggregate monitoring.



Figure 6: CTA bus in service on Halsted Street in Chicago, IL (source: Chicago Transit Authority).

6.2.1. Scenario A: route-level monitoring (Route 8)

In the first scenario, we focus on a single route (Halsted 8), representing a monitoring task at the corridor-level. The CTA Route 8 Halsted bus line serves as a major north–south transit corridor within the city of Chicago and offers daily service with extended operating hours designed to accommodate a wide range of commuter schedules. The route operates between Halsted/79th and Broadway/Waveland, with weekday service beginning at 4:05 a.m. and ending at 12:30 a.m., and similar spans on weekends adjusted by direction of travel (Chicago Transit Authority, 2026). The high intensity of the service, sustained demand, and exposure to localized disruptions make Route 8 a good candidate to study demand monitoring at the corridor level.

Figure 7 shows the total monthly ridership observed for Route 8. The series shows strong seasonality, gradual long-term change, and abrupt drops associated with major external events. A road map of Route 8 portion is provided in Fig. 8 to give spatial context of the route.

6.2.2. Scenario B: system-level monitoring (aggregate average per route)

In the second scenario, we consider system-wide monitoring. A direct aggregation that sums ridership across all routes can be misleading because some new routes are added, while some are retired from the system at different time points. This may cause changes in the aggregate to reflect shifts in network coverage rather than changes in underlying demand. To address this issue, we construct an aggregate series defined as

$$x_t = \frac{\sum_r \texttt{MonthlyTotal}_{r,t}}{N_t}, \tag{11}$$

where N_t denotes the number of routes that report data in month t. This normalization produces a quantity interpretable as the average monthly ridership per active route.

The resulting aggregate series is shown in Fig. 9. This scenario represents a system-wide monitoring task aimed at tracking overall demand conditions while remaining robust to route turnover.

6.3. Data Exploration and Preprocessing

The exploratory plots indicate that both series are seasonal and nonstationary. The histograms in Figure 10 are skewed and heavy tailed, and the month-of-year violin plots (Figure 11) confirm the systematic variation within a year. Therefore, we apply the monitoring on residuals rather than on raw counts. Similar approach has been used by other studies (for example see Zeroual et al. (2017)).

For each series x_t, we form a baseline forecast $\hat{x}_t$ using a seasonal naïve model:

$$\hat{x}_t = x_{t-12}, \tag{12}$$

with a fallback to a 36 month trailing mean (shifted by one period) when seasonal lags are unavailable. Residuals are defined as

$$e_t = x_t - \hat{x}_t \tag{13}$$

The raw ridership series for Route 8 and the system-level aggregate are shown in Figures 7 and 9. Because residual variance changes over time, we standardize using a trailing rolling z score with window $w = 36$:

$$z_t = \frac{e_t - \mu_{t-1}^{(w)}}{\sigma_{t-1}^{(w)}}, \tag{14}$$

where $\mu_{t-1}^{(w)}$ and $\sigma_{t-1}^{(w)}$ are computed from the past residuals only. The resulting standardized residuals appeared more stable (Figure 10).

Figure 12 presents the monitoring results for CTA Route 8 using the EWMA chart, the CUSUM chart and the proposed RL-based EWMA chart applied to the standardized residual series. We note two major disruptions that are visible across all methods. These are the demand shift around 2007–2008 and the abrupt collapse in early 2020 associated with the COVID-19 pandemic. These events provide natural reference points for evaluating how each chart responds to both moderate change and shock. As observed in the plots, all three approaches detect these disruptions. However, their signaling patterns differ markedly in timing, duration, and persistence.

During the 2007–2008 period, the EWMA and CUSUM charts show a gradual accumulation of deviations, crossing their respective thresholds after deviations persist for several months and then remained in an

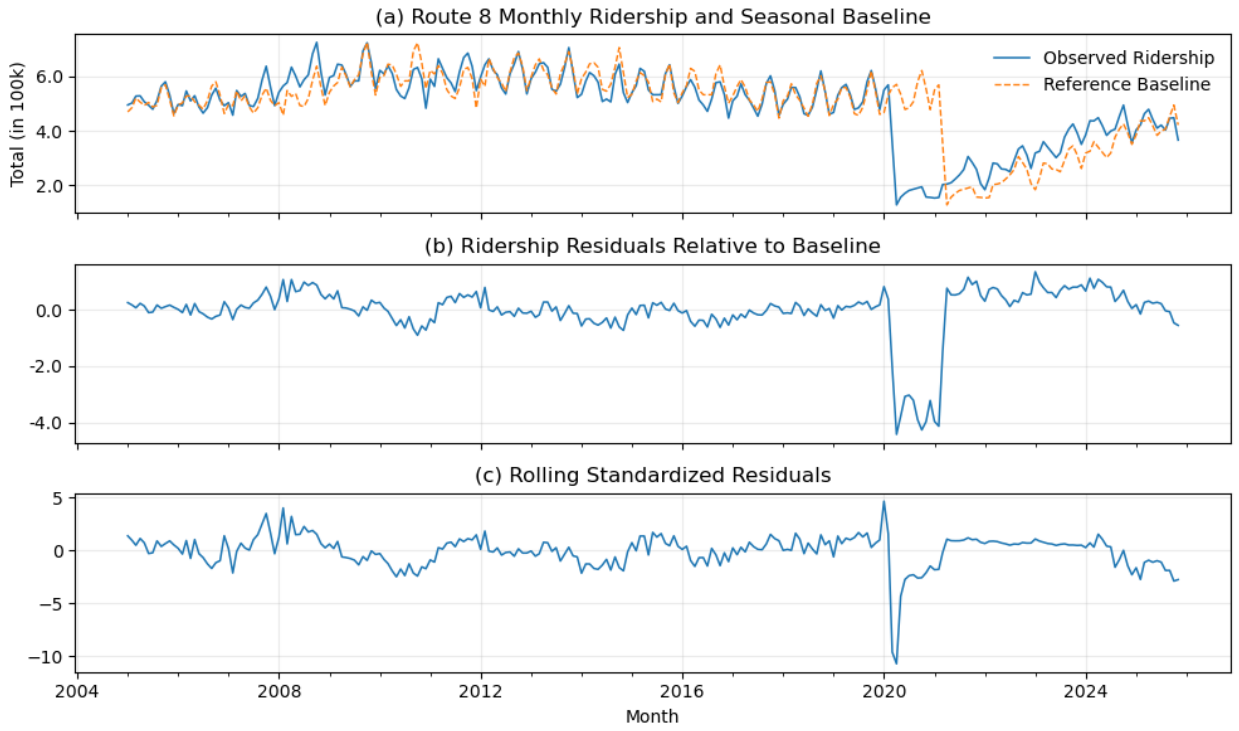


Figure 7: Monthly ridership for CTA Route 8 and successive stages of preprocessing: (a) observed ridership with seasonal reference baseline, (b) deviations from the baseline, and (c) rolling standardized residuals used as input for monitoring.

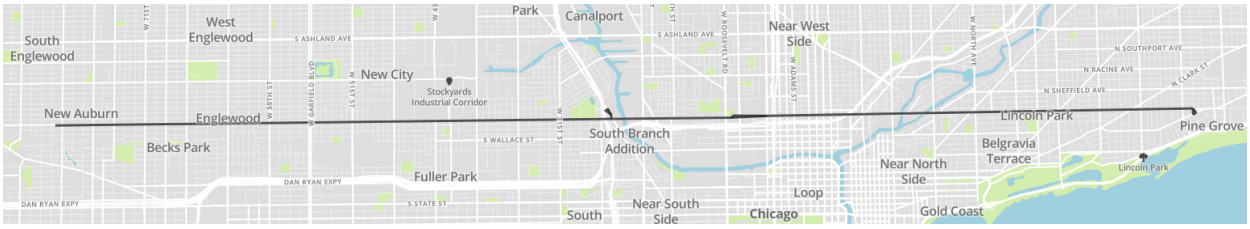


Figure 8: Road map portion of CTA Route 8, Halsted.

OOC state for an extended duration. This behavior reflects the sensitivity of the classical charts to sustained shifts. The proposed chart also responds to this shift, but over a shorter period. The probability of signaling increases sharply near the onset of the change and then returns to low levels once the deviation stabilizes. This contrast becomes even more visible during the COVID-19 shock. The magnitude of the collapse drives both EWMA and CUSUM very far from their control limits, which results in a prolonged OOC state.

Outside of these two major events, differences in sensitivity remain. It can be seen from the plots that the EWMA and CUSUM charts occasionally accumulate deviations during moderate changes, leading to an OOC state even in the absence of clearly defined changes. By contrast, the RL-based chart remains largely IC during such periods, with only minor and transient increases in signaling probability. From these results, it can be deduced that, while classical charts emphasize sustained deviation and tend to signal continuously once triggered, the proposed chart prioritizes the detection of regime transitions. From an operational perspective, this distinction is important as concise alerts concentrated around the change points may be more actionable than persistent alarms for extended periods.

Figure 13 presents another monitoring results (Scenario B) for the aggregate average per route series applied to the same three competing charts. As with the previous results, all three charts respond strongly to the major system-wide disruptions around 2007–2008 and

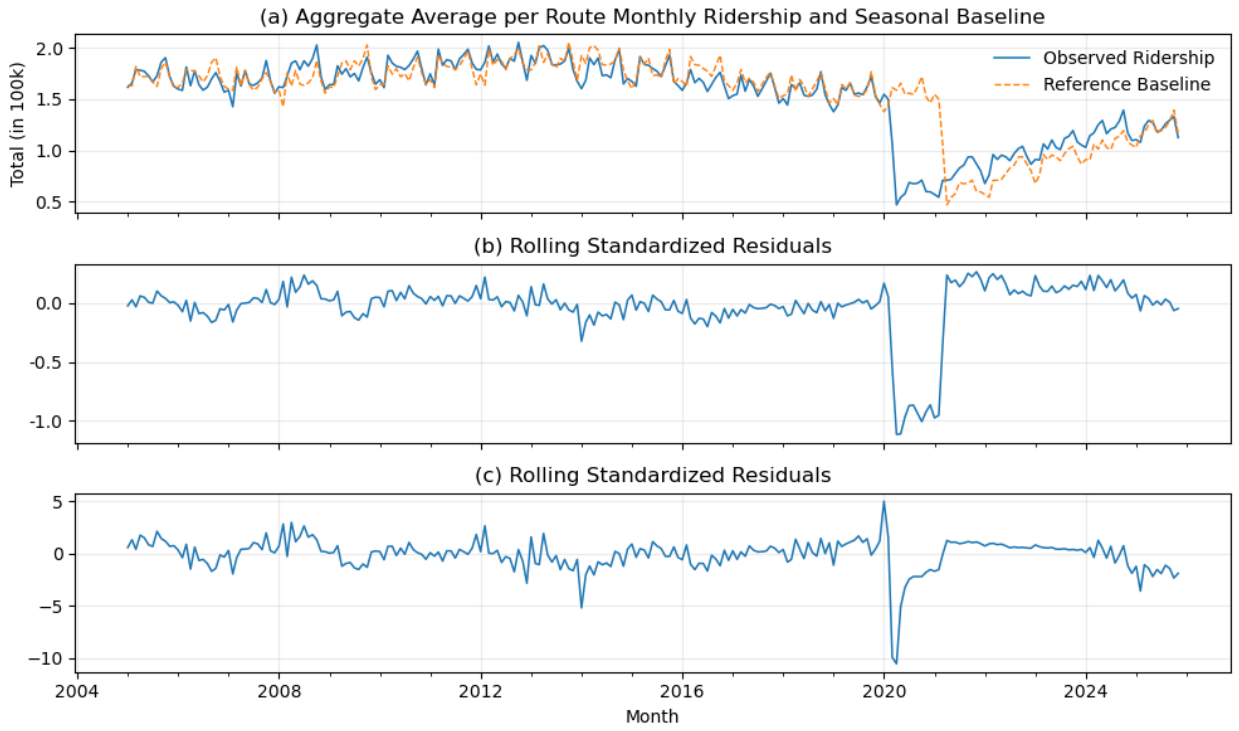


Figure 9: Aggregate average per route monthly ridership and successive stages of preprocessing: (a) observed ridership with seasonal reference baseline, (b) baseline adjusted deviations, and (c) rolling standardized residuals used as input for monitoring.

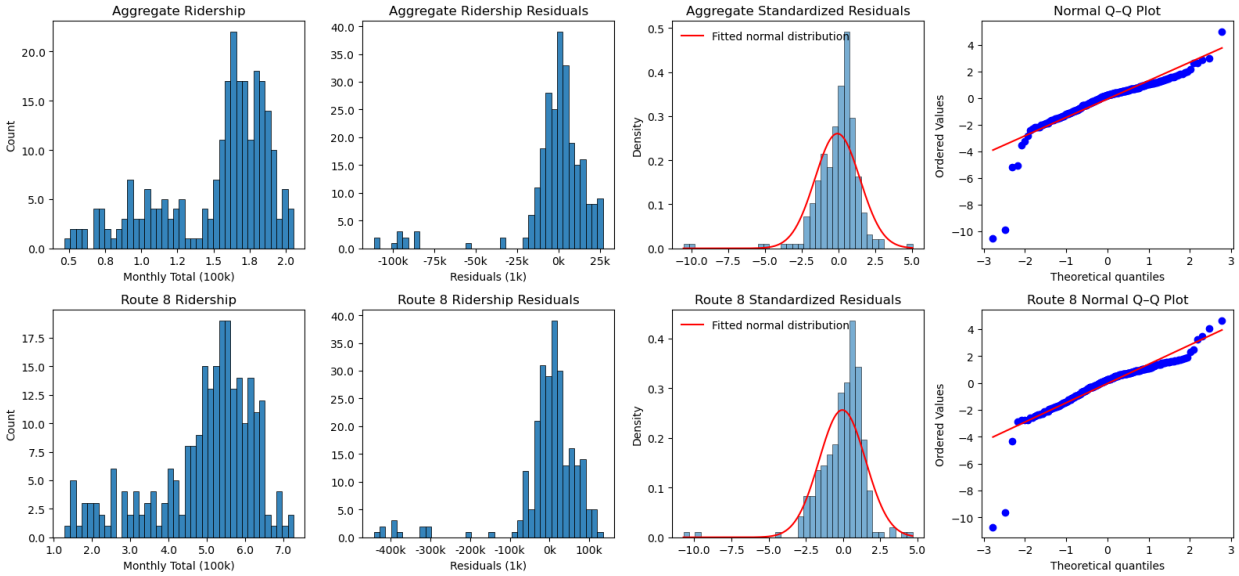


Figure 10: Aggregate and Route-Level Ridership Distributions After Baseline Adjustment (Distributions of raw ridership, baseline adjusted residuals, and rolling standardized residuals for the aggregate average per route series (top) and Route 8 (bottom), together with normal Q–Q plots illustrating departures from normality.

the sharp collapse in early 2020.

Again, the classical EWMA and CUSUM charts show sustained deviations once their thresholds are crossed, especially during the COVID-19 period, where

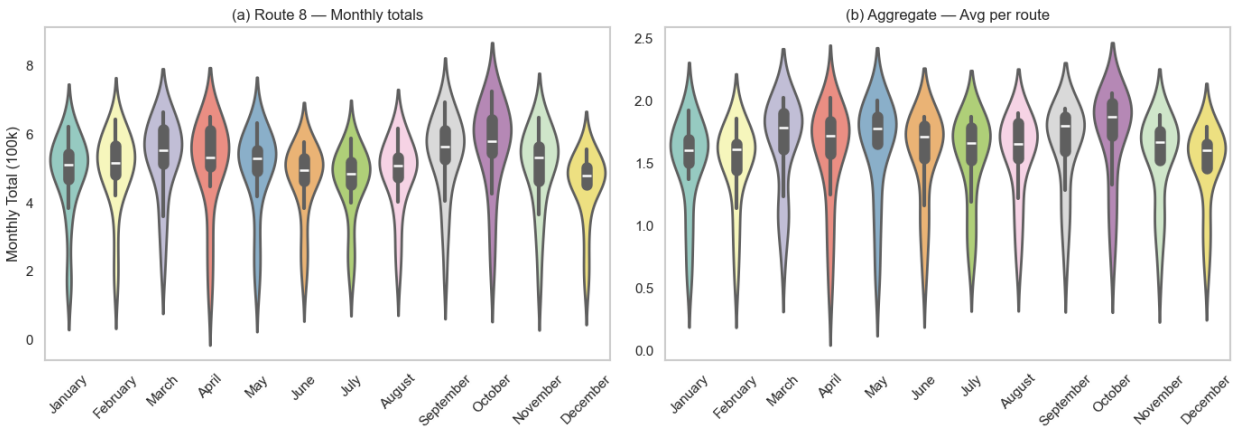


Figure 11: Month-of-year violin plots for Route 8 (a) and the aggregate series (b).

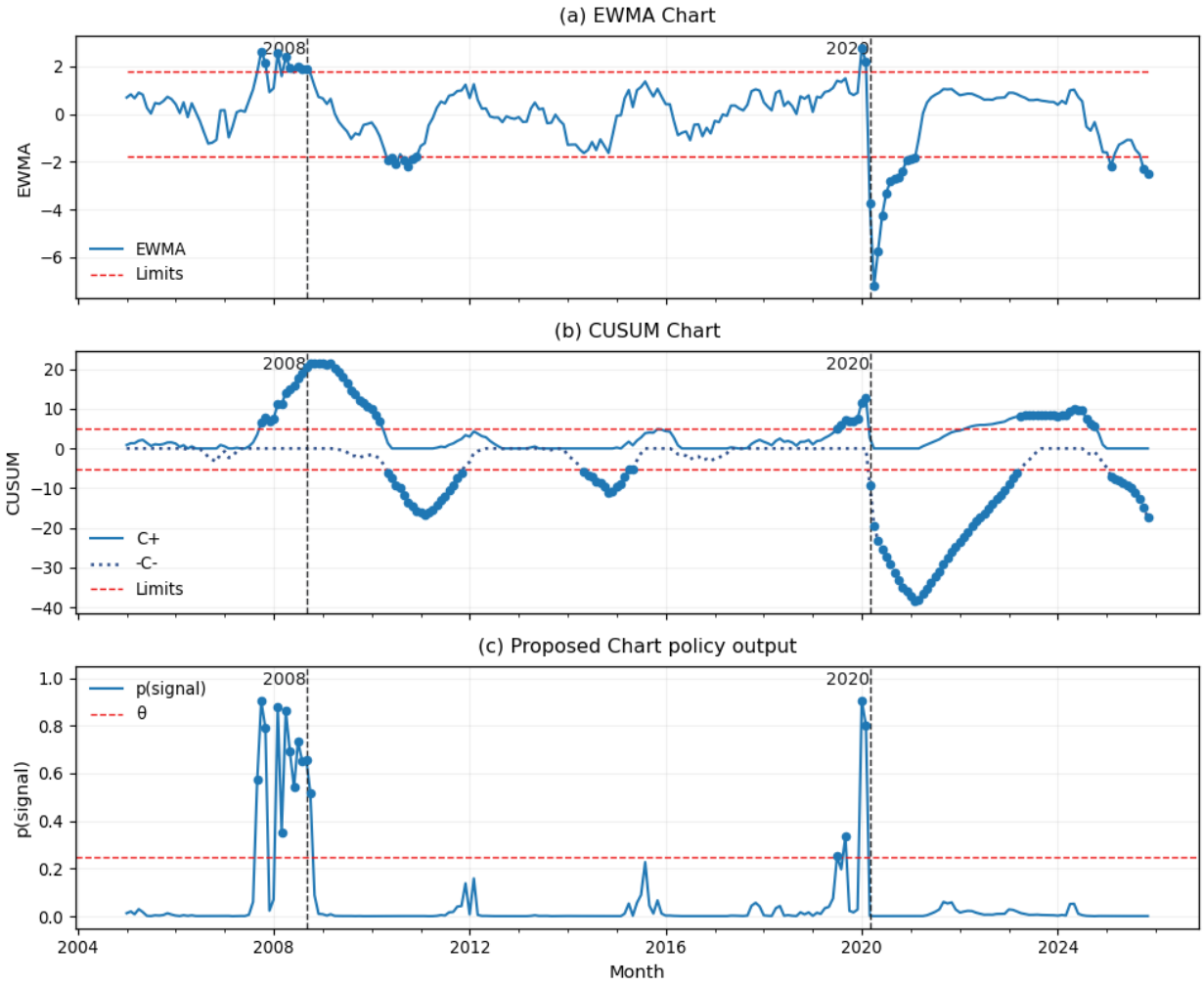


Figure 12: Monitoring results for CTA Route 8: (a) EWMA chart, (b) CUSUM chart, (c) proposed RL-based EWMA chart.

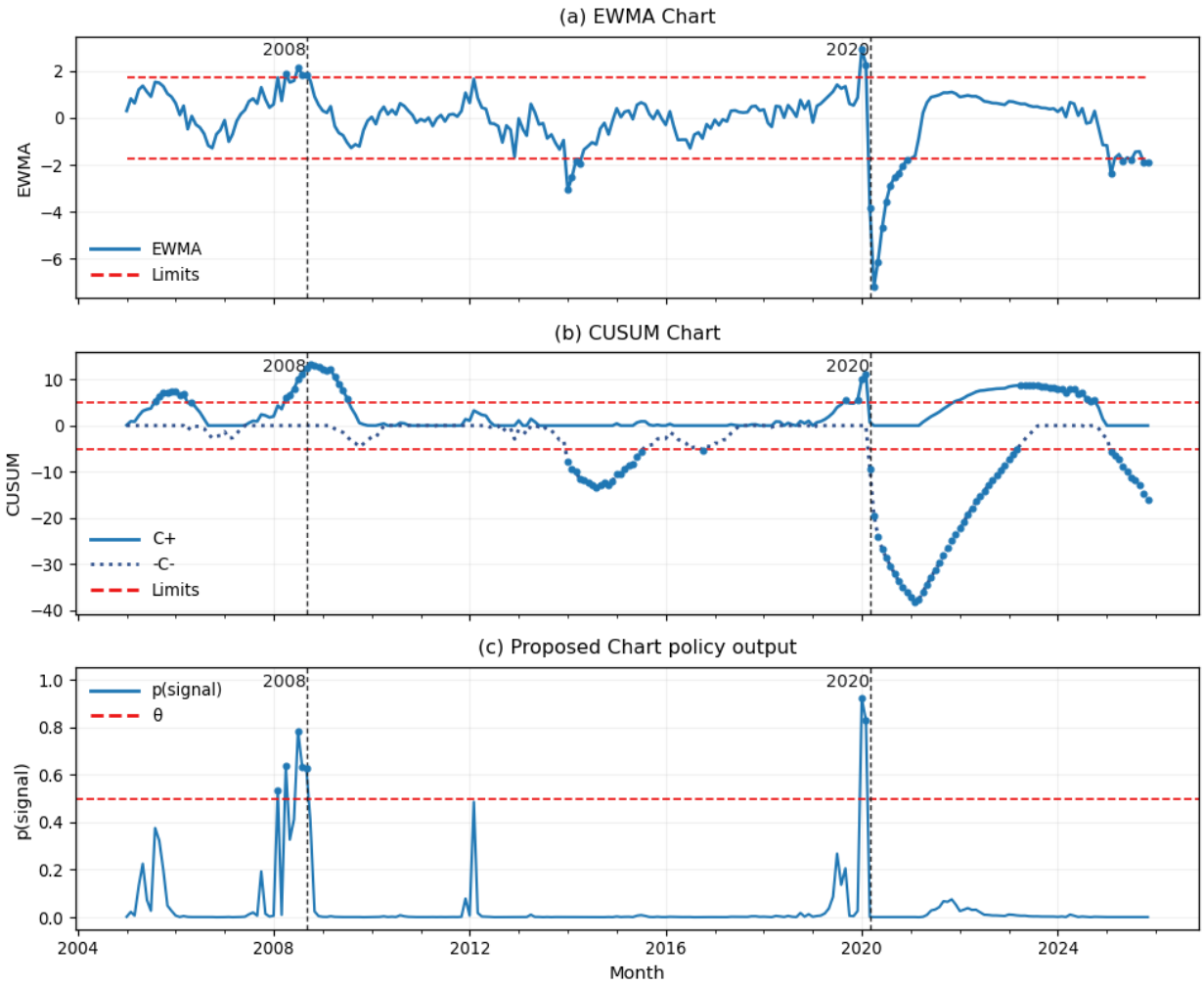


Figure 13: Monitoring results for the system-level aggregate (average monthly ridership per active route): (a) EWMA chart, (b) CUSUM chart, (c) proposed RL-based EWMA chart.

the magnitude of the disruption causes a prolonged OOC state throughout much of the recovery phase. In contrast, the proposed chart produces short, concentrated spikes in signaling probability near the onset of each disruption, and then rapidly returns to low values. Similar to Scenario A, the proposed chart remains largely IC outside the major disruptions. In contrast, the classical charts occasionally produce signals as random deviations accumulate.

6.4. Managerial Insights

This paper proposes a new design of EWMA control chart so that it detects changes and gives an alarm at the right time for real operations. The model output stays simple, which is either to continue with the current process or stop and check, but the decision rule comes from simulated process data and then calibrated so the false alarm rate meets a predefined target. For managers, the implication is fewer false alarms and faster detection of real shifts, without manual trial and error on chart constants.

7. Conclusion

This study introduced an RL-based EWMA control chart in which the signaling rule is learned directly from data and subsequently calibrated to meet standard in-control performance requirements. The proposed scheme formulates the signaling decision as a sequential decision-making problem and employs reinforcement learning to map the EWMA statistic to a stopping

action, while a post training calibration step is used to enforce the desired ARL_0. The performance of the chart was evaluated using *ARL*, EQL, and CED.

Comparisons with adaptive EWMA, classical EWMA and CUSUM charts indicate that the proposed scheme achieves improved ARL_1 performance for small and moderate shifts, and lower EQL values. The results show that learning a data-driven signaling policy can provide faster detection than the existing methods.

We also present a case study that illustrates how the proposed RL-based EWMA control chart can be applied to real transit ridership data to identify statistically significant deviations from typical demand patterns. The results show that the proposed chart produces concise signals concentrated around major demand shifts. These findings suggest that the proposed framework can serve as a practical decision support tool to monitor transportation ridership under nonstationary and shock prone conditions.

Some simplifying assumptions were adopted in this work. In particular, the process distribution was assumed to be normal with known parameters, shifts were modeled as permanent changes in the mean, and the EWMA smoothing parameter was fixed during training and evaluation. We investigate the effect of parameter estimation in the case that they are not known. Future research could extend this framework in several directions. Potential topics may include investigating non-Gaussian or dependent data, extending the method to multivariate monitoring problems, and allowing the RL agent to jointly adapt both the signaling rule and chart parameters. Exploring alternative reward structures and online learning schemes may also provide insight into how RL can be integrated more broadly into statistical process monitoring. Economic design in which the cost parameters decide independently the *ARL* values without post training calibration can also be studied.

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8. Appendix

Table 8: PPO training, environment, and reward settings used for the RL-based EWMA control chart.

Item	Value
RL library / policy	Stable-Baselines3 2.3.0, PPO with `MlpPolicy`
Learning rate	3×10^{-4}
Discount factor γ	0.99
GAE parameter λ_{GAE}	0.95
PPO clip range ϵ	0.1
Value-function clip range	0.1
Entropy coefficient	1×10^{-5}
Batch size	128
n_steps	2048
PPO epochs per update (n_epochs)	10
Number of environments	8
Total timesteps	2×10^5
Observation normalization	`VecNormalize`: norm_obs=True, norm_reward=False, clip_obs=10.0
In-control distribution	$X_t \sim \mathcal{N}(0, 1)$
EWMA statistic	$Z_t = \lambda X_t + (1 - \lambda) Z_{t-1}$, $Z_0 = 0$; observation is (Z_t, X_t)
Action space	0=WAIT, 1=SIGNAL
Training shift injection	Geometric, $p = 0.02$ per step
Training shift magnitude	$\delta \sim [0.5, 3.0]$ chosen uniformly;
Episode truncation	MAX_STEPS=10,000
Reward: delay cost c_d	0.2
Reward: signal reward R_{sig}	1
c_s	0
Penalty for false alarm P_{FA}	1
Unlisted PPO parameters	Stable-Baselines3 2.3.0 defaults